ODOT RESILIENCE IMPROVEMENT PLAN

Ohio Department of Transportation

October 2024

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Introduction

Extreme weather events and natural disasters have cost Ohio billions of dollars, and these events are only expected to worsen in the future.[1] Between 1980 and 2023, Ohio has experienced 90 declared disasters that caused losses of over $1 billion.[2] The costly impacts of extreme weather events and natural disasters on Ohio's transportation system—from deteriorating pavement to washing out culverts to damaging bridges—can increase maintenance, operational, rehabilitation, and replacement costs. As extreme weather events and natural disasters become more frequent and intense, the Ohio Department of Transportation (ODOT) needs to increasingly prioritize resilience to ready itself for future impacts to the transportation system.

Defining Resilience

Resilience is "the ability to prepare for changing conditions and withstand, respond to, and recover rapidly from disruptions."[3] A resilient transportation system can adapt to, withstand, and recover from intense weather and natural disasters.[4]

The overarching goal of the Resilience Improvement Plan (RIP) is to increase the resilience of the Ohio transportation system.

Resilience is central to ODOT's mission of providing a transportation system that is safe, accessible, well maintained, and positioned for the future. A safe transportation system is one that can withstand extreme weather events and natural disasters. An accessible and well-maintained transportation system is one that can recover quickly and efficiently from extreme events and natural disasters. A transportation system that is positioned for the future is one that is designed and managed with extreme weather events and natural disasters in mind and prepared to adapt. For ODOT's transportation system to be safe, accessible, well maintained, and well positioned for the future, it needs to be resilient.

1 NOAA National Centers for Environmental Information. 2022. Ohio State Climate Summary. https://statesummaries.ncics.org/downloads/Ohio-StateClimateSummary2022.pdf

2 NOAA National Centers for Environmental Information. 2023. US Billion-Dollar Weather and Climate Disasters. https://www.ncei.noaa.gov/access/billions/

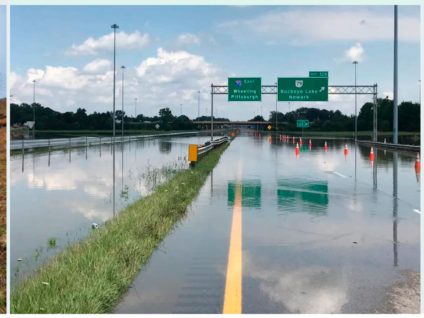
3 FHWA. November 2015. "Transportation System Resilience to Extreme Weather and Climate Change." Accessed on December 7, 2022. https://ops.fhwa.dot.gov/publications/fhwahop15024/index.htm

4 ODOT. October 2018. System Resiliency. https://dam.assets.ohio.gov/image/upload/transportation.ohio.gov/Programs/AccessOhio/White%20Papers/AO45_WhitePaper_SystemResiliency.pdf

Examples of Natural Hazard Impacts in Ohio

Landslides are a common issue in parts of Ohio and repairs can be very costly and often take several months to complete.[5]

I-70 near Buckeye Lake sees widespread major flooding during extreme precipitation events. In 2020, the Licking River hit record levels and completely inundated I-70.[6]



Route 7 has seen multiple rockslides and landslides that have caused significant damage to the roadway and resulted in closures for multiple days.[7]

Goals and Objectives

The RIP aims to streamline ODOT's efforts to integrate resilience into standard decision-making processes to ensure the continued reliability and safety of the transportation network.

The primary objectives of the RIP are to:

- Identify and prioritize locations for targeted resilience investments.
- Identify opportunities to further embed resilience into long-range transportation planning and decision-making at ODOT.
- Encourage a culture of resilience at ODOT using vulnerability or risk data and other resilience practices to regularly inform decisions that support a safe and reliable transportation network.

5 ODOT. 2021. ODOT working to prevent costly landslides and rockslides. https://www.transportation.ohio.gov/about-us/news/statewide/odot-working-to-prevent-costly-landslides-and-rockslides

6 Newman, J. 2020. "I-70 by Buckeye Lake Flooded River Hits Record Levels of Flooding." https://www.sciotopost.com/70-buckeye-lake-flooded-river-hits-record-levels-flooding/

7 Taylor, A. 2018. "Ohio rockslide blocks Route 7, road expected to be closed multiple days." https://wchstv.com/news/local/rockslide-blocks-route-7-in-ohio

The RIP supports the goals of Ohio's *Access Ohio 2045* Long-Range Transportation Plan of Safety, Efficiency & Reliability, Preservation, Economic Competitiveness, Quality of Life, and Environmental Stewardship. The plan also directly supports the following goal objectives:

SAFETY	Support effective response to and recovery from natural disasters, emergencies, and incidents.
PRESERVATION	Maintain transportation assets in a state of good repair.
EFFICIENCY & RELIABILITY	Increase the efficiency and reliability of moving people and freight.

These priorities are emphasized throughout this document, with the sections below further detailing how the implementation of the RIP will help advance ODOT's progress on these goals.

This plan summarizes the systemic approach that ODOT used to assess the vulnerability of the highway system infrastructure, provides a list of prioritized resilience projects, and outlines the steps to implement resilience practices. The core elements of this plan include a summary of the risk-based vulnerability assessment, the resilience investment plan listing priority resilience investments, and a description of the systemic approach to implementing resilience in practice. Additional RIP sections provide context and scope of the RIP, and Appendices provide supporting information. See Figure 1.

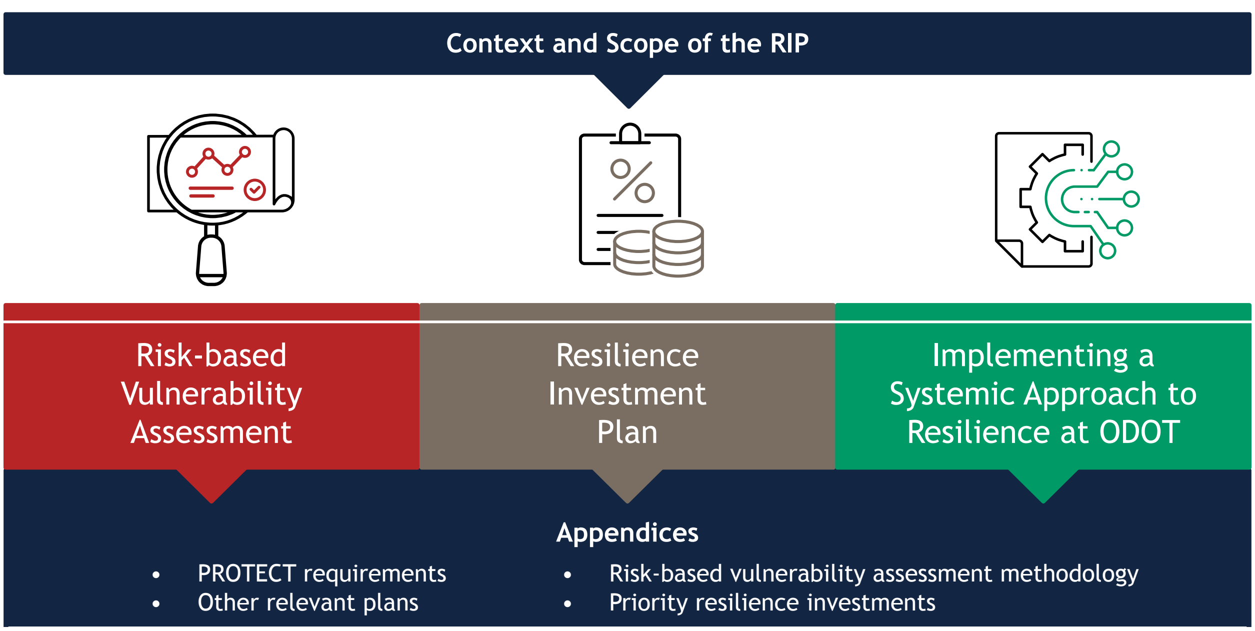


Figure 1. Summary of the components of the RIP.

Context and Scope of RIP

Ohio's RIP was developed to meet the requirements outlined in the Federal Highway Administration's (FHWA) Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Formula Program guidance.[8] As outlined in the Bipartisan Infrastructure Law (BIL), state departments of transportation or metropolitan planning organizations (MPOs) that develop RIPs that meet these requirements reduce the non-federal cost share for projects by 7%. An additional 3% will be reduced if the RIP is incorporated into the statewide long-range transportation plan. Appendix 1 details the required elements of a RIP and which sections of Ohio's RIP address each of the PROTECT requirements.

How This Plan Will Increase Resilience

The RIP will increase the resilience of the transportation system by identifying ODOT's greatest risks and vulnerabilities in conjunction with building on existing resilience practices across the organization. ODOT will be able to prioritize locations for resilience investments by understanding which assets are most vulnerable to weather events and natural disasters, which is a critical step in further increasing the resilience of the transportation system. The RIP will also build on existing resilience efforts across ODOT and identify opportunities to enhance resilience and facilitate its integration into standard decision-making processes across ODOT.

Scope of the Resilience Improvement Plan

Timeframe

The RIP evaluates present and future climate scenarios to align with long-range plans (e.g., *Access Ohio 2045*), investment decisions, and the lifespan of infrastructure assets. Existing state plans (e.g., the Transportation Asset Management Plan [TAMP]) and upcoming projects are incorporated as relevant. The focus of the RIP is to identify the near-term (1 to 5 years) actions that ODOT can take to increase the resilience of the transportation infrastructure that is most vulnerable while providing a long-term roadmap for investment decisions. By considering both immediate and long-range planning activities and investments, the RIP ensures that ODOT's approach to resilience is effective and sustainable.

Geographic Scale

The RIP is statewide and is focused on **roads and bridges** as the core components of ODOT's surface transportation system, which are critical to system performance and characterize the largest share of ODOT's investment in the transportation system. The bridges included are all those in the National Bridge Inventory, which includes bridges and culverts greater than 20 feet. The road sections are based on road segments in Ohio's Linear Referencing System (of any ownership or functional class).

8 FHWA. 2022, July 29. Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Formula Program Implementation Guidance. https://www.fhwa.dot.gov/environment/sustainability/resilience/policy_and_guidance/protect_formula.pdf

Weather Events and Natural Disasters Considered

The RIP focuses on **flooding and geohazards (landslides and rockfalls)** because these hazards pose the greatest risk to ODOT's infrastructure and assets. Higher frequency and intensity of precipitation events are the driving factors increasing risks for both flooding and geohazard events. Potential flooding impacts range from increased scour action at bridge piers and abutments to compromised pavement integrity on roads constructed on expandable clay soils. Potential geohazard impacts include slope erosion, slumping of ditches, and reduced pavement subgrade stability. Other potential hazards, such as extreme heat and high winds, were not included in the risk-based vulnerability assessment as they pose less risk than flooding and geohazards based on observed climate trends and climate projections. However, strategies to address these hazards are not precluded from ODOT's resilience strategies. Future iterations of the RIP could extend the risk assessment to other hazards and asset types, such as culverts, conduits, and/or ports.

Critical Interdependencies Across Sectors

ODOT recognizes that **the transportation system is just one component of community resilience** and relies, in part, on other sectors to provide reliable and safe transportation to all Ohioans.

For example, many aspects of the transportation system depend on energy and telecommunications. During extreme weather events and natural disasters, there can be significant service disruptions or suspensions. When there is flooding or a geohazard impeding access to energy or telecommunications, repairs following extreme events can be delayed. This can lead to more widespread and longer lasting power outages.

Other sectors also depend equally on the transportation system. Emergency response services, for example, depend on the accessibility and reliability of the transportation network to quickly reach injured people and critical facilities following extreme events and natural disasters.

As climate change worsens extreme weather events and natural disasters in Ohio, it is essential that ODOT's transportation system is a robust one to minimize cascading impacts across other sectors.

Connection to Existing Plans

The RIP was developed as an integral part of ODOT's transportation planning process, and as such, is consistent with and complementary to existing ODOT plans, as well as state and local hazard mitigation plans. By coordinating with other planning activities, the RIP ensures a systemic approach to transportation system resilience across Ohio.

Table 1 outlines how the RIP builds on existing plans such as Ohio's Long-Range Transportation Plan *(Access Ohio 2045)*, ODOT's TAMP, ODOT's State Freight Plan *(Transport Ohio)*, and state and local hazard mitigation plans. Appendix 2: Existing ODOT Plans describes existing ODOT plans in more detail.

Table 1. ODOT Plans Related to the RIP

Plan	Relation to the RIP
Access Ohio 2045 (2020)	The RIP is aligned with the goals and strategies outlined in *Access Ohio 2045*, including the identification of transportation assets that are most vulnerable to extreme weather events.
TAMP (2022)	As described in the TAMP, the RIP builds on existing resilience work done to date in the Infrastructure Resiliency Plan and the Vulnerability Assessment Scoring Tool (VAST). The TAMP summarizes all ODOT resilience work completed or planned to date, which is also described in the RIP.
Transport Ohio (July 2022)	As ODOT's State Freight Plan, *Transport Ohio* calls for the development of an Infrastructure Resiliency Plan that will identify and prioritize critical multimodal infrastructure. The RIP will meet this objective and help Ohio prepare for more frequent extreme weather events.
Hazard Mitigation Plan (2019)	Ohio's State Hazard Mitigation Plan tasks ODOT with developing a risk management plan to address risks, including those associated with climate change and extreme weather events. The RIP addresses the climate hazards that are of greatest concern to the transportation sector—flooding and geohazards.

Other State and Local Plans

HAZARD MITIGATION PLANS

The RIP is consistent with Ohio state and local hazard mitigation plans. All 88 counties in Ohio develop a local hazard mitigation plan (LHMP) that is updated every 5 years to meet Federal Emergency Management Agency (FEMA) criteria for LHMPs. Each plan presents the county's long-term strategy to reduce losses due to extreme weather events and natural disasters and ultimately increase community resilience. Hazard identification and risk data from each local plan is tracked on the Ohio Emergency Management Agency's State Hazard Analysis Resource and Planning Portal[9] and incorporated into the state hazard mitigation plan at least every 5 years.[10]

Ohio's State Hazard Mitigation Plan (HMP) identifies hazards that pose the greatest threat to people and infrastructure in the state, evaluates emerging risks, and provides a blueprint of mitigation actions that state agencies can take to reduce risk. The plan emphasizes that climate change is a looming threat that will amplify extreme weather events and identifies flooding and drought, from increasingly variable precipitation and longer heat waves, as the main climate threats that Ohio will face in the coming years. The HMP includes a literature review of projected impacts due to climate change and of adaptation planning done within Ohio to date. Hazards analyzed include flooding, tornadoes, winter storms, landslides, wildfire, dam and levee failure, coastal flooding, earthquakes, coastal erosion, severe summer storms, invasive species, and land subsidence. The plan provides an overview of each hazard, the probability of local hazard events, and an estimate of the potential dollar losses to state assets located in hazard areas. Flooding, landslides, and winter storms are specifically identified as threats to Ohio's transportation system in the 2019 statewide HMP. The HMP recommends that ODOT install sensors to monitor roadway flooding, incorporate weather resilience into the Traffic Operation Assessment Systems Tool, and develop a risk management plan to address climate hazards that could damage highway condition and performance.

CITY AND MPO PLANS

The RIP builds on what Ohio's cities and counties have already accomplished to integrate climate change and resilience into decision-making processes. The three largest cities in Ohio—Columbus, Cleveland, and Cincinnati—have completed climate action and sustainability plans.[11,12,13] Regional organizations, such as the Northeast Ohio Areawide Coordinating Agency, are including climate resilience in investment planning.

ODOT is coordinating with the 17 MPOs and six regional transportation planning organizations (RTPOs) in Ohio to develop the RIP and implement its recommendations. ODOT reviewed the MPO and RTPO priorities to incorporate them wherever possible.

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9 Local Plan HIRA & Vulnerability Analysis. https://services.dps.ohio.gov/MIP/Reports/Reports/GetLHMPHIRAAndVulnerability

10 State of Ohio Enhanced Hazard Mitigation Plan. 2019. https://www.ema.ohio.gov/static/mip/links/2019_sohmp-FullCopy.pdf

11 City of Columbus. 2021, December. Columbus Climate Action Plan. https://www.columbus.gov/Services/Public-Utilities/About-Public-Utilities/Office-of-Sustainability/Columbus-Climate-Action-Plan

12 Sustainable Cleveland. 2018. The Cleveland Climate Action Plan: 2018 Update. https://www.sustainablecleveland.org/climate_action

13 City of Cincinnati. 2023, April. Green Cincinnati Plan. https://www.cincinnati-oh.gov/oes/climate/climate-protection-green-cincinnati-plan/

Risk-Based Vulnerability Assessment

As part of the development of this RIP, ODOT conducted a risk-based vulnerability assessment to help prioritize vulnerable assets and locations to inform the development of solutions. This vulnerability assessment is built on data and methods from ODOT's VAST, which has been updated annually since 2019. The data and methods in VAST were enhanced with additional metrics to quantify risk and inform resilience investment needs.

This section describes the methodology and results of ODOT's risk-based vulnerability assessment. Figure 2 provides an overview of the coverage of the assessment.

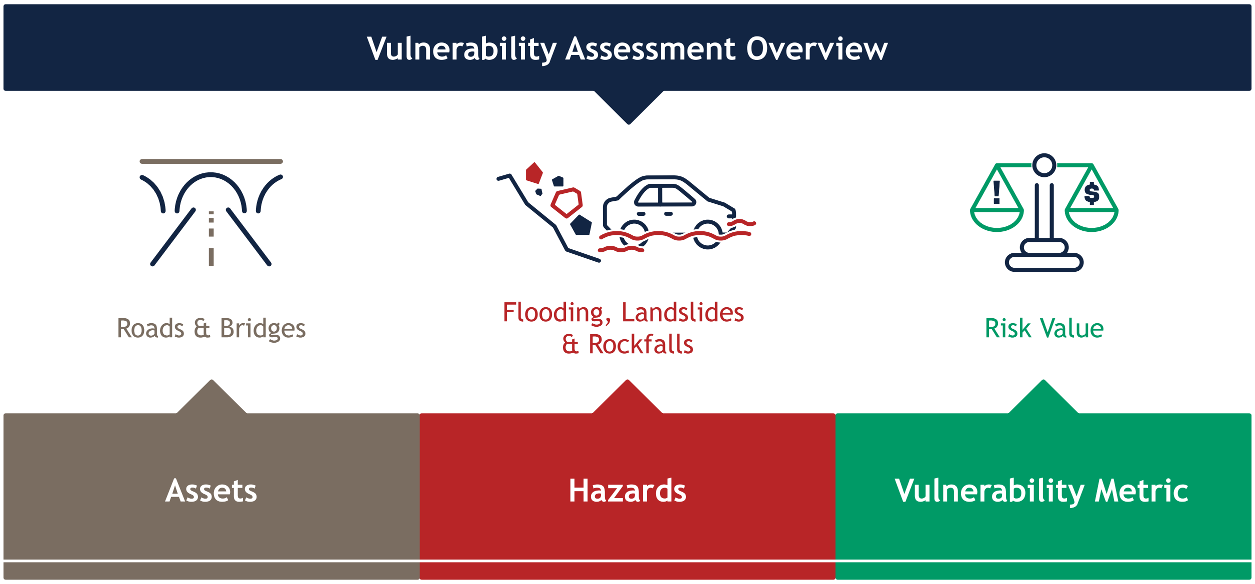


Figure 2. Assets, hazards, and metrics covered in ODOT's vulnerability assessment.

Methods

The risk-based vulnerability assessment quantifies the average annual cost of risk (risk value) of a range of climate and natural hazards. The cost of risk is assessed at the individual asset level (e.g., individual bridges and road segments) and rolled up to evaluate risk at multiple scales, such as corridors, ODOT districts, and statewide. Note that although the vulnerability assessment results provide a useful framework for quantifying flooding and landslide risks to individual assets, the analysis is not intended to be overly predictive and will not capture all risks associated with either hazard. For example, the analysis cannot capture all flood risks like a hydrologic and hydraulic model could. The analysis provides a useful starting point for understanding the relative risk cost for ODOT's assets and prioritizing adaptation investments, but additional analyses or modeling efforts may be needed to refine the results.

Assets Analyzed

The assessment covers roads and bridges and includes any road segments in Ohio's Linear Referencing System (of any ownership or functional class), filtered to those that had at least a 0.25-mile contiguous section or the entire segment within the FEMA 100-year floodplain.[14] In total, the assessment includes 5,120 road segments, totaling more than 1,300 miles of roadway.

14 The FEMA 100-year floodplain corresponds to a historical 1% annual chance of flooding. In this report, 'floodplain' refers to the FEMA 100-year floodplain.

Bridges included in the assessment come from the National Bridge Inventory for Ohio, which includes all bridges and culverts that have a span of at least 20 feet. This assessment includes the 4,729 structures on state-maintained roadways and within the FEMA 100-year floodplain.

The assessment is currently limited to assets in the FEMA 100-year floodplain, however, ODOT is aware that there are flooding and geohazard risks outside of the floodplain boundaries. Future enhancements to the risk assessment could expand coverage to areas outside the FEMA floodplain.

Climate Hazards and Scenarios Analyzed

The assessment analyzes risks from flooding, landslides, and rockfalls. These hazards were chosen because they pose the greatest risk to ODOT's roads and bridges. Additionally, these hazards are expected to worsen under future climate change. Extreme precipitation events have been increasing in Ohio since the mid-1990s, and these events are expected to become more intense and frequent in the future.[15] This will increase the risk of flooding across Ohio, which can cause significant damage to ODOT's roads and bridges. Furthermore, the most frequent and damaging landslides and rockfalls are typically caused by prolonged or heavy rainfall.[16]

Vulnerability Metrics and Outputs Used

The assessment computes a quantitative **risk value** for each asset, the expected average annual cost of risk to each asset.

For each asset, **risk value** is calculated in dollars, as illustrated in Figure 3.

Risk Value ($) = Threat Likelihood (%) x Consequence ($)

Cumulative across hazards (flooding, landslides, and rockfalls)

Figure 3. Risk value calculation.

Risk value represents the average annual economic cost of a threat or hazard impacting a bridge or road segment. This metric helps ODOT understand the greatest risks from a monetary perspective and provides initial data on the benefits of improving a given asset.

Risk value is a product of both the *likelihood* of a hazard and the *consequences* of the hazard occurring. For example, risk value is low where there is no threat or where there is a threat but with limited consequences, such as a road that is flooded frequently but not significantly damaged, and high where the likelihood of a hazard occurring and economic damage as a result of the hazard are high.

15 NOAA National Centers for Environmental Information. 2022. Ohio State Climate Summary. https://statesummaries.ncics.org/downloads/Ohio-StateClimateSummary2022.pdf

16 US Geological Survey, Department of the Interior. 2018. Overview of Rainfall-Induced Landslides. https://www.usgs.gov/programs/landslide-hazards/science/overview-rainfall-induced-landslides

Consideration of Probability/Likelihood

Threat likelihood is intended to capture the average likelihood that a given asset would be closed due to flooding or geohazards in a given year. The **threat likelihood** captures the cumulative probability of closure from the following types of events:

- Flooding – overtopping only (i.e., water over the roadway that results in relatively short-term closures until the water recedes)
- Flooding – damage (i.e., flooding that causes damage to the roadway infrastructure)
- Landslide
- Rockfall

The severity of a hazard will dictate the amount of time that a roadway or bridge is closed for repairs. In the worst-case scenario, the asset will need to be replaced entirely. This is more likely for landslides or rockfalls and less likely for assets affected by flooding. The vulnerability assessment methodology attempts to account for the annual probability of damage for each asset and hazard.

Appendix 3 summarizes key assumptions that drive the estimated annual likelihood of closure for each asset. Closure duration and damage level assumptions by hazard are also included in the appendix. These assumptions factor into the consequence calculations described in the next section.

Consequence

Consequence is estimated in dollar terms based on both the potential **owner cost** (i.e., the cost of asset repair or replacement) and the **user cost** of disruptions. Consequence cost estimates are high level and should only be used to support the prioritization of vulnerable assets.

OWNER COST

Owner cost is the cost to repair or replace a damaged road segment or bridge and is based on factors such as the asset's size and typical asset replacement or repair costs (Figure 4).

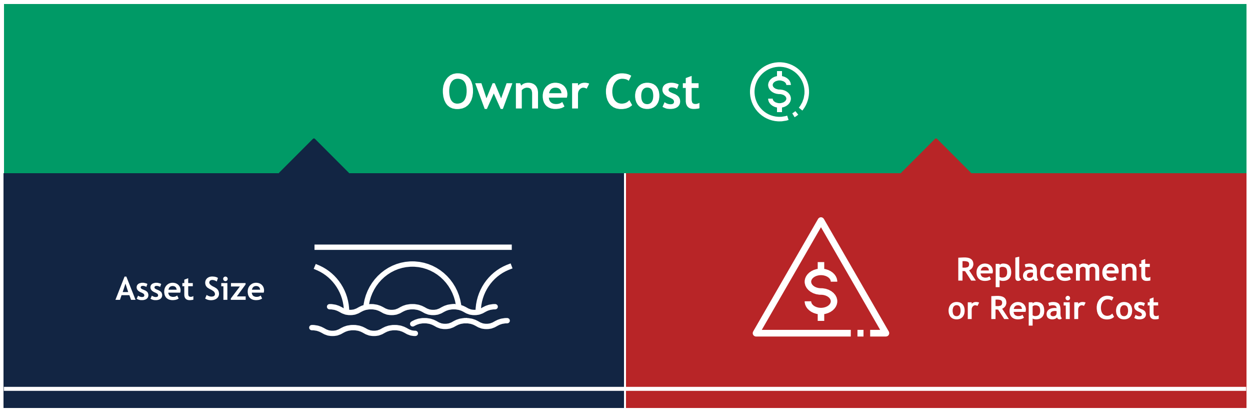


Figure 4. Summary of owner cost inputs. Red denotes a constant.

USER COST

User Cost is the added cost to drivers and society if drivers are forced to take an alternative route due to road or bridge closure or lose access to a destination. Taking a longer route to avoid a road closure results in additional travel time and added vehicle operating costs. Overall user costs are a function of how far users have to go out of their way to avoid a closed segment (and how long it takes) and how many people are affected, as summarized in Figure 5.

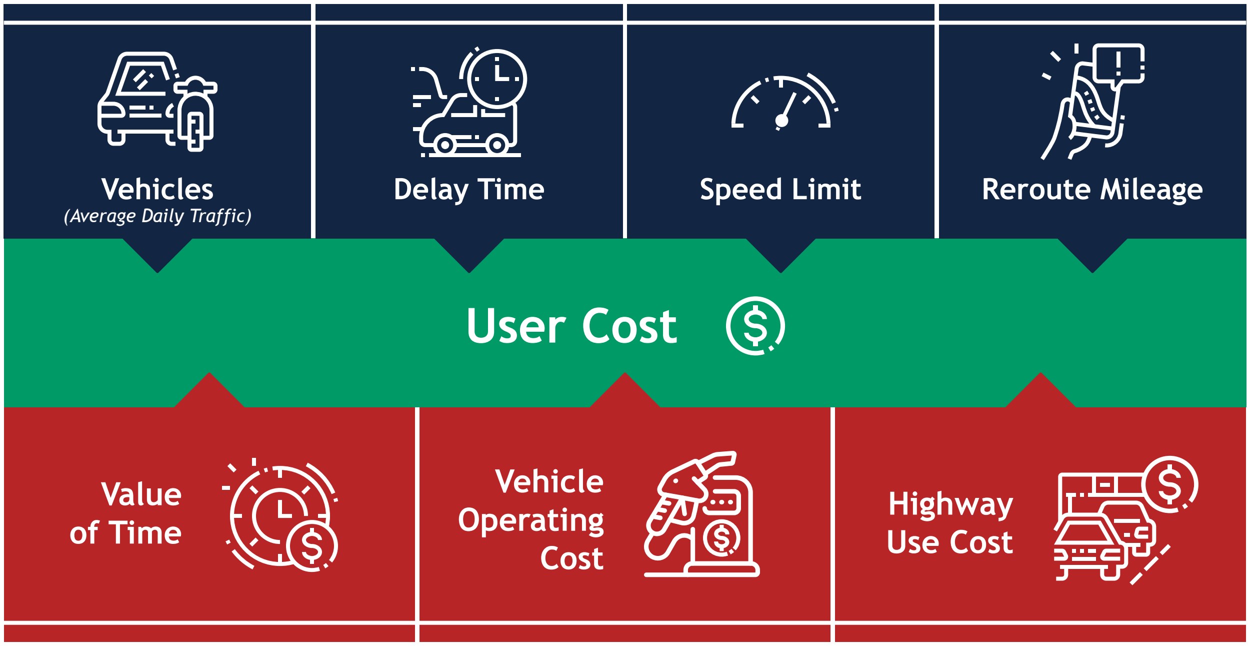


Figure 5. Summary of user cost inputs. Red denotes a constant.

User cost has two components: the cost resulting from a **full closure** and the cost resulting from a **partial closure** *(impeded flow cost)*.

FULL CLOSURE USER COST

Full closure user cost incorporates:

- **Detour length/system redundancy.** The additional distance and time traveled in the event of a road segment or bridge closure. See Appendix 3 for methodology on calculating detour length.
- **Volume** of passenger vehicles and trucks affected.
- **Economic value.** Hourly or per mile cost of lost time, highway use, and vehicle operations. This analysis uses nationwide values provided by FHWA.

The total daily user costs are then multiplied by the estimated closure duration for each hazard.

PARTIAL CLOSURE USER COST

Partial closure user cost, or impeded flow cost, is calculated by estimating the added length of time it would take a user to traverse a segment based on an assumed reduced travel speed. The same added costs of time are incurred without added highway use costs since there is no added driving distance. The total daily costs of partial closures are then multiplied by the duration of partial closures for each hazard.

Figure 6 summarizes the inputs that contribute to the overall consequence value for each asset.

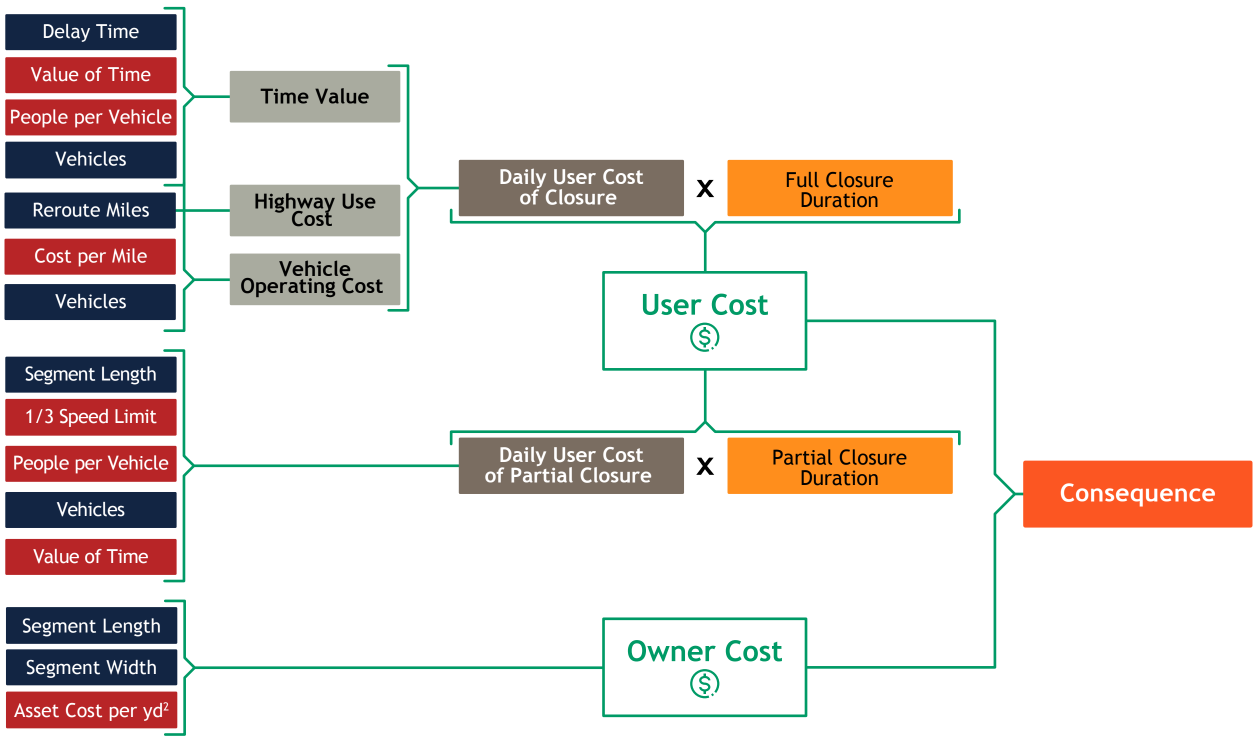


Figure 6. Consequence methodology summary. Blue boxes indicate variables and red boxes indicate constants.

Additional details on the risk assessment methods are included in Appendix 3.

Results

The risk assessment reveals that the annual risk value due to flooding and geohazards totals more than $113 million per year, mostly driven by the risk to roads. District 10 has the greatest risk value of any district with 34% of the statewide total, driven by rockfall hazards, landslides, and flooding (see Table 2). Rockfalls are the hazard with the greatest risk value, driven mostly by the extended duration of disruptions and associated user costs. Damage-related flooding costs are lower overall versus flood events without damage given the significantly lower likelihood of damaging flood events.

Table 2. Total Risk Value by District and Hazard (Roads and Bridges)

District	Flooding Risk Value (No Damage)	Flooding Risk Value (Damage)	Landslides Risk Value	Rockfall Risk Value	Total Risk Value
1	$2,626,116	$1,790,073	$313,821	$85,886	$4,815,896
2	$1,393,230	$1,068,048	$195,434	$117,055	$2,773,767
3	$1,119,711	$384,030	$8,992	$101,260	$1,613,994
4	$632,288	$170,397	$12,854	$261,261	$1,076,800
5	$6,741,533	$2,876,221	$253,672	$1,710,208	$11,581,633
6	$1,331,095	$437,968	$15,913	$356,174	$2,141,150
7	$1,653,193	$389,306	$14,574	$315,559	$2,372,632
8	$2,929,040	$595,726	$9,138,063	$740,259	$13,403,088
9	$4,459,989	$2,841,664	$5,153,415	$11,655,446	$24,110,515
10	$3,645,267	$1,932,030	$12,059,160	$21,001,772	$38,638,229
11	$1,804,686	$531,868	$1,806,265	$6,515,676	$10,658,495
12	$508,656	$105,828	$14,679	$36,071	$665,234
Total	$28,844,803	$13,123,159	$28,986,841	$42,896,627	$113,851,430

Figure 7 and Figure 8 show maps of all roads and bridges analyzed, respectively, symbolized by total risk value across hazards. Figure 9 and Figure 10 isolate the top 20 highest risk roads and bridges, respectively. The majority of the highest risk roads are in Southeast Ohio.

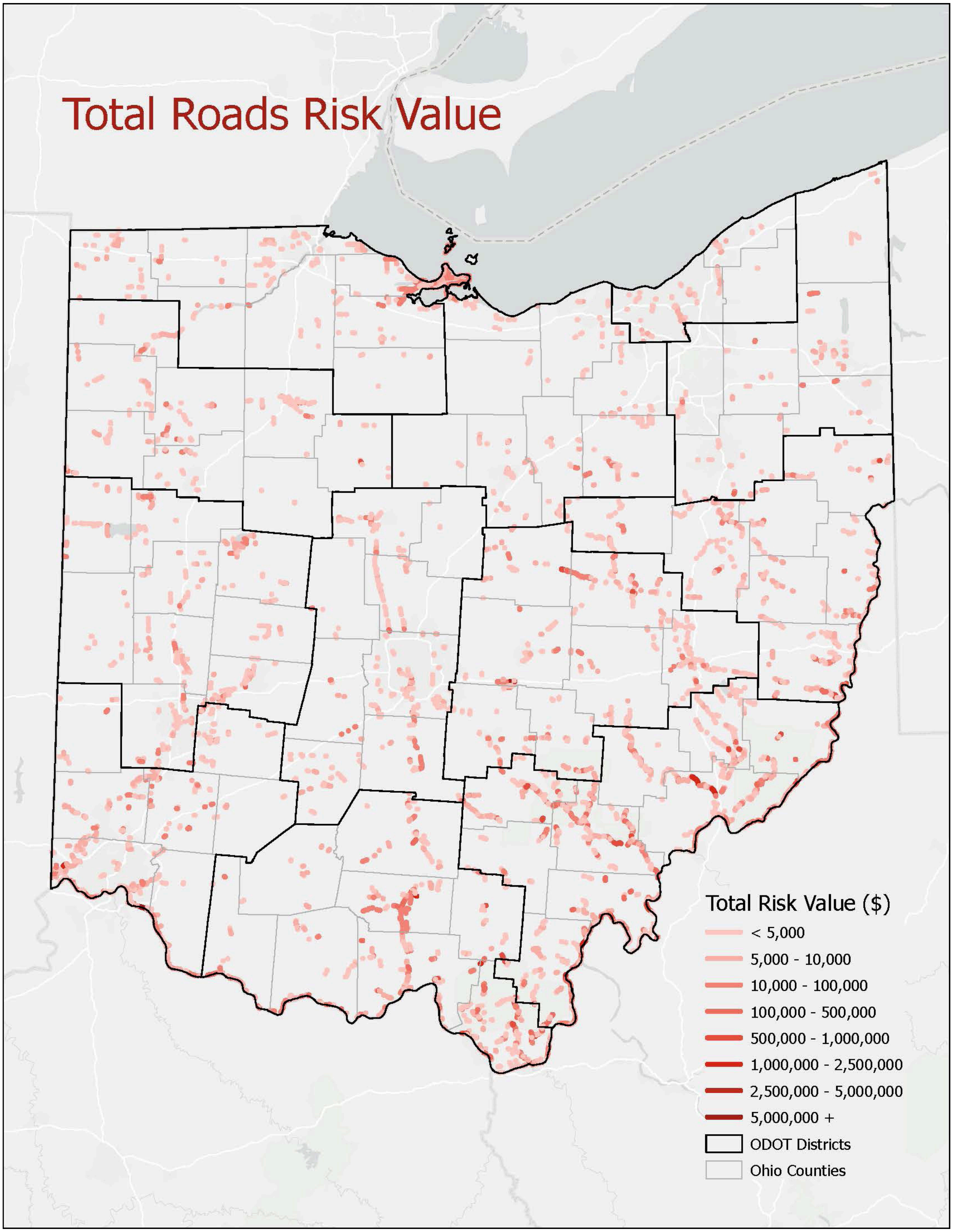


Figure 7. Roads risk value, all segments.

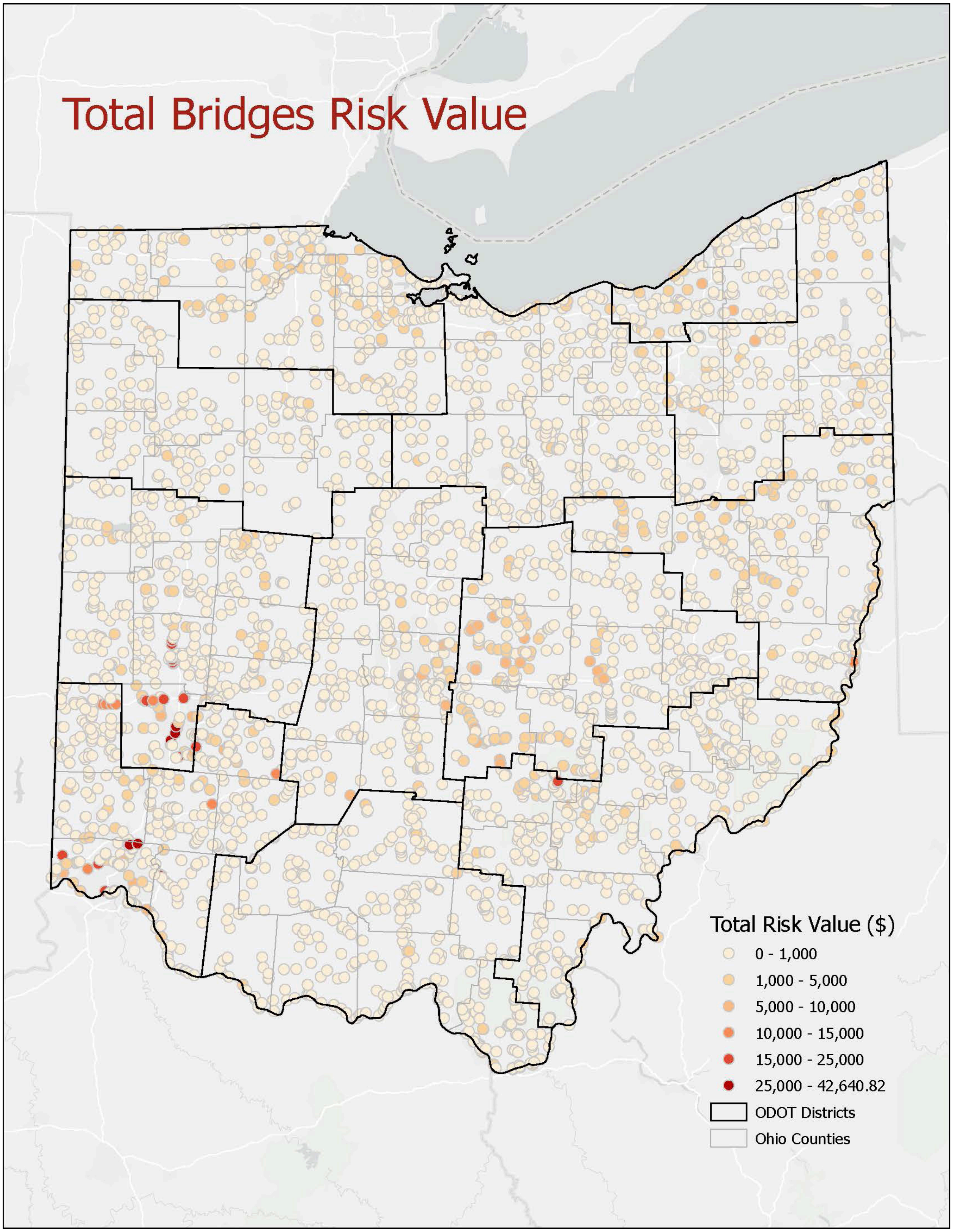


Figure 8. Bridges risk value, all assets.

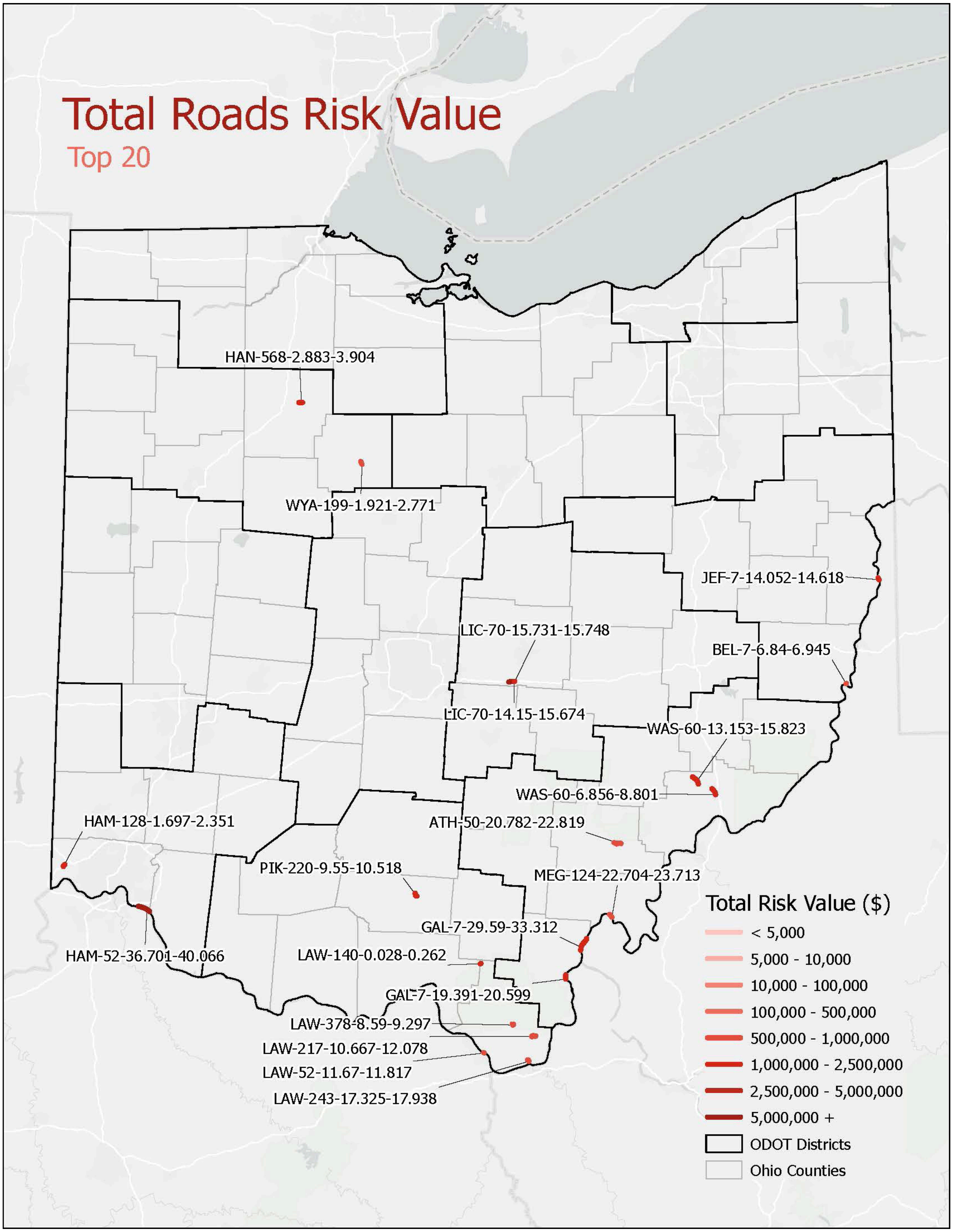


Figure 9. Top 20 highest risk road segments, by risk value.

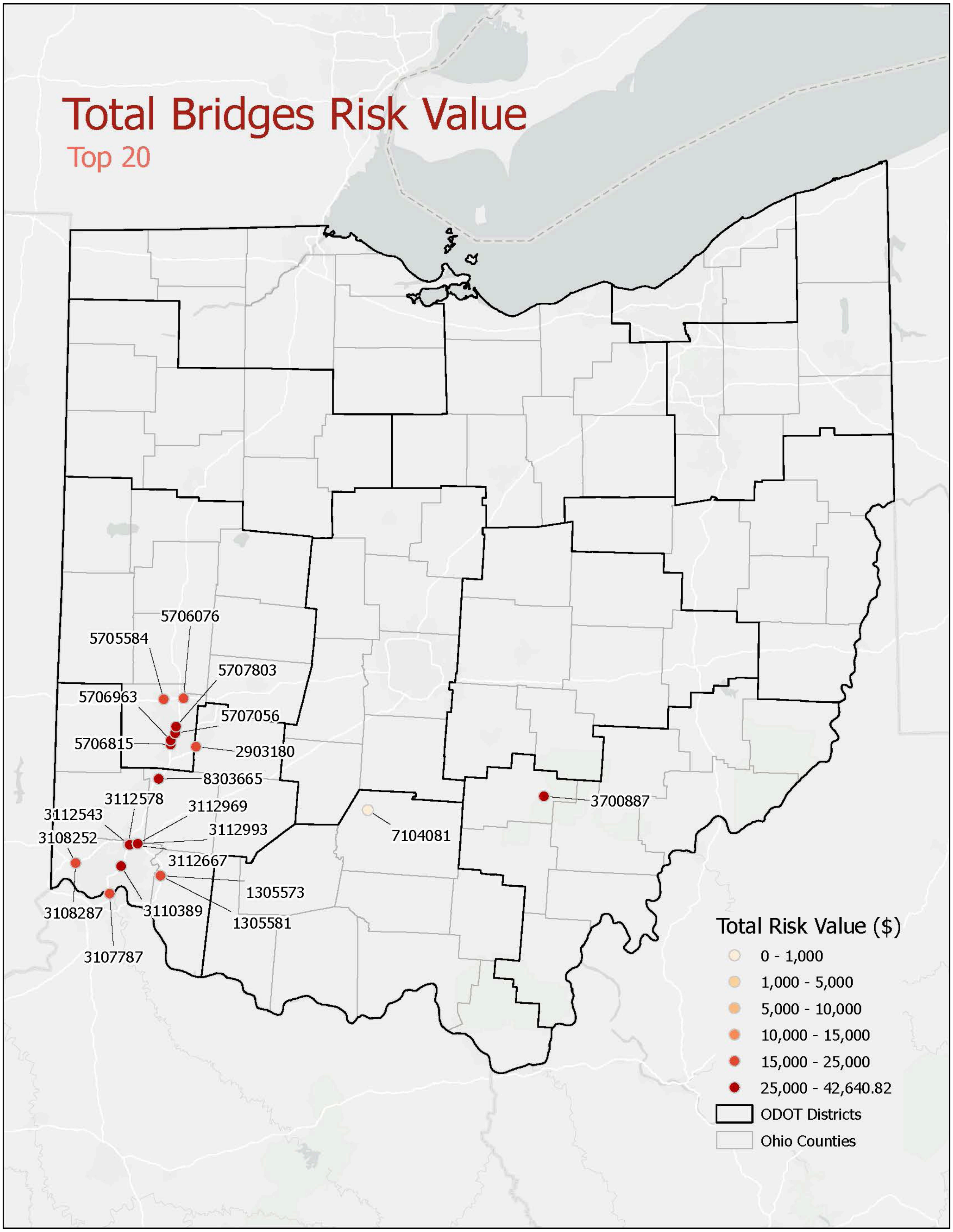


Figure 10. Top 20 highest risk bridges, by risk value.

Figure 11 illustrates the total risk value across roads and bridges by hazard, disaggregated by owner risk value and user risk value. Owner risk value represents the majority of the risk value at more than $59 million per year. The total average annual user cost is estimated at more than $51 million per year.

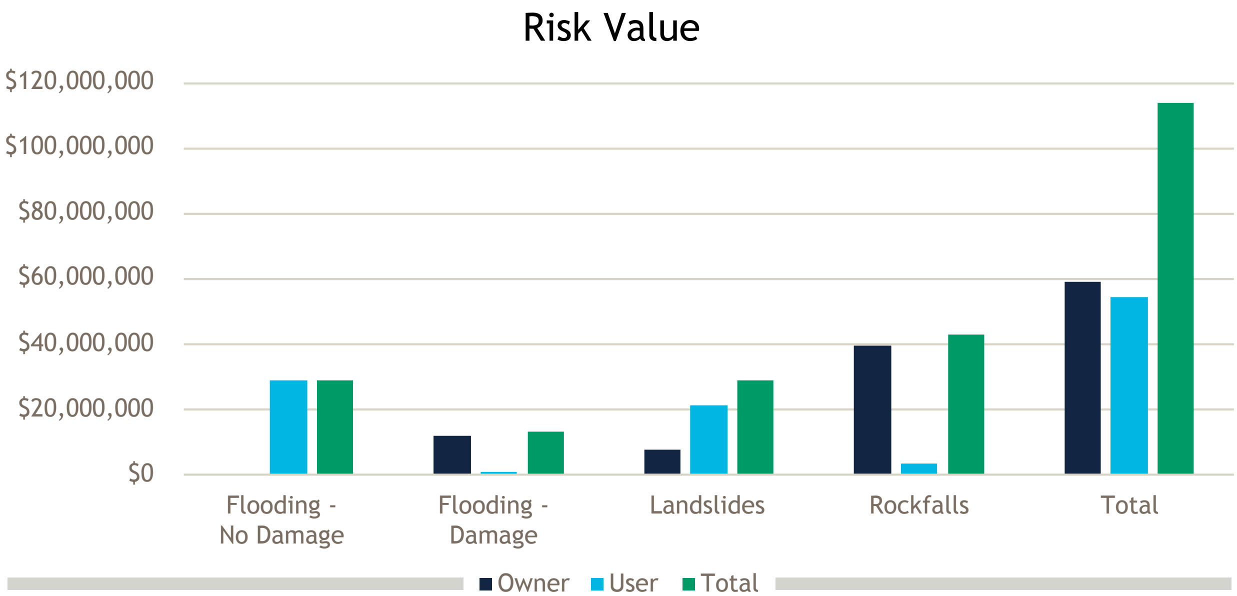


Figure 11. Total risk value by hazard.

Corridor Risks

The vulnerability assessment also identifies whether individual at-risk segments are grouped in corridors that could warrant collective risk management strategies at the corridor or even watershed scale, as opposed to at the asset level. The highest risk corridors include the following (log points are approximate):

- LIC 70 (14.15-15.748)
- US 52 across Southern Ohio (ADA-52-10.87-11.23, 18.53-19.51, 23.01-23.05; BRO-52-1.72-3.98, 8.17-8.94; CLE-52-2.87-6.54; HAM-52-36.70-40.07; LAW-52-2.90-4.32, 11.67-14.00)
- State Route (SR) 7 along the Ohio River
- WAS SR 60 (6.43-21.345)
- MEG SR 124 (22.04-29.526, 53.042-55.722. 66.154-66.262)
- HAN SR 568 (2.88-3.904)
- LAW SR 775 (8.036-10.624, 23.598-23.959)
- GAL SR 218 (0.876-3.186, 5.017-5.459)

Summary of Implementation

ODOT used the results of the risk-based vulnerability assessment and other factors to prioritize projects in the resilience investment plan and will continue to use them to inform ongoing project programming and prioritization decisions. Although the monetized risk value captures many of the important consequences of potential transportation system disruptions, it does not monetize all potential social consequences, such as loss of access to critical destinations or the impacts of transportation system disruptions on social equity. Other factors were calculated and considered alongside the risk values to prioritize potential investments, including the following:

Distance to critical facilities. Each asset is scored based on its proximity to critical facilities that provide or maintain essential services, such as utility operations centers, shelters, hospitals, and fire stations.

Social equity impacts. This element indicates which assets would have the greatest impact on disadvantaged communities within the state if disrupted. To measure this, the assessment applies the “social vulnerability” component rating from the US Department of Transportation’s Equitable Transportation Community Explorer.[17] In this index, each census tract receives a score on a scale of 0 to 100 based on the following indicators:

Percentage of

- the population age 65+
- the population age 17 or younger
- the population with a disability
- the population (age 5+) with limited English proficiency
- the population that is uninsured
- the population with an income below 200% of the poverty level
- people age 25+ with less than a high school diploma
- people age 16+ who are unemployed
- total housing units that are renter occupied
- occupied houses that spend 30% or more of their income on housing (less than $75k income)
- households with no internet subscription
- total housing units that are mobile homes
- Gini Index, which is a summary measure of income inequality

17 US DOT. 2023. USDOT Equitable Transportation Community (ETC) Explorer. https://experience.arcgis.com/experience/0920984aa80a4362b8778d779b090723/page/Understanding-the-Data/

Resilience Investment Plan

ODOT used the results of the risk-based vulnerability assessment described above, along with input from district staff and headquarters departments, to develop a priority list of resilience investment projects.

Appendix 4 provides the full list of the potential priority projects, listed in alphabetical order by county. Social and economic considerations are reflected here via both the scoring methodology (e.g., detour length) and staff input on critical sites.

Consideration of Natural Infrastructure

ODOT is considering natural infrastructure, through the use of nature-based solutions, as described in the Implementing a Systemic Approach to Resilience at ODOT section. In addition, the agency is developing a set of resilience strategies that can be applied in common situations (e.g., flooding, landslides). To accelerate the application of these strategies, 5 to 10 project cut sheets are being developed for specific locations with known high vulnerability to flooding or geohazards.

PROTECT Funding and Matching

The federal funding for the PROTECT program is separated into formula and discretionary. The formula funding is a set amount provided to each state, while the discretionary funding is available nationally and allocated based on competitive grants. Within the state of Ohio, the PROTECT formula funds are received by ODOT and are programmed through the Division of Planning. ODOT will primarily match the PROTECT program funding with state funds. PROTECT formula funds started in 2022 and are programmed annually over the immediate 5 years, starting in 2022.

Implementing a Systemic Approach to Resilience at ODOT

Strategies to continuously enhance ODOT's resilience to weather events and natural disasters are embedded in every part of the organization. Furthermore, ODOT is dedicated to building on existing resilience efforts and is striving toward the full integration of resilience into standard decision-making processes to ensure the continued reliability and safety of the transportation network. From asset management to operations and maintenance to emergency preparedness, ODOT prioritizes resilience and continues to explore new opportunities to increase resilience at both the asset and system levels.

An overview of ODOT's policy and regulatory framework for resilience and existing resilience practices (with a focus on asset management, design, operations and maintenance, and emergency preparedness) is found in this section. Figure 12 summarizes the systemic approach described in this section and illustrates how resilience is considered across every part of ODOT. See the sections below for more information.

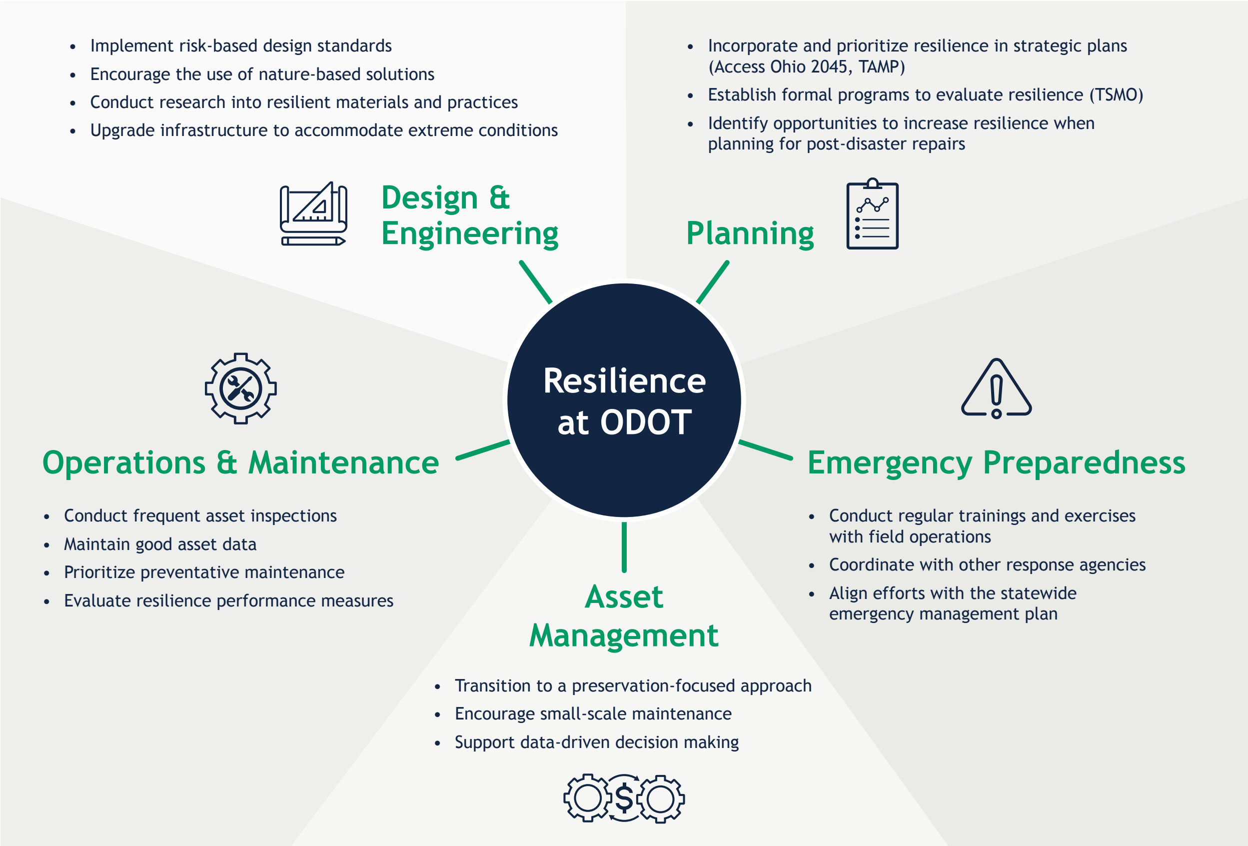


Figure 12. ODOT's systemic approach considers resilience across every part of the organization.

ODOT's Culture of Continuous Improvement

ODOT maintains a strong culture of continuous improvement as an organization. Our staff are empowered and motivated to continue exploring new and innovative ways to further increase our resilience, from conducting new research projects and studies to testing alternative materials and design methods. At ODOT, we are never satisfied. We are constantly searching for ways to plan, design, and build better and stronger.

As climate, extreme weather, and other conditions change, our culture is a key enabling factor to our ability to adapt to change and remain resilient over time.

ODOT Organizational Overview

Resilience principles are integrated across ODOT. Within ODOT's Central Office there are 13 divisions that support Ohio's 12 regional ODOT districts by providing guidance, policies, and support. The Operations, Engineering, and Planning divisions are responsible for asset management, design, and maintenance, as well as emergency preparedness. These divisions spearhead the bulk of the resilience work at ODOT.

ODOT's Policy and Regulatory Framework for Resilience

ODOT has several policies and frameworks in place to instill resilience practices throughout the organization. Resilience is incorporated into many of ODOT's strategic plans, which direct ODOT's overall vision of a world-class transportation system that improves the lives of all Ohioans.

Access Ohio 2045, ODOT's **Long-Range Transportation Plan**, provides a strategic blueprint to guide ODOT's transportation policies and investments. One of the goals articulated in this plan is to improve the resilience of the transportation network to increasingly frequent and extreme weather events.

The **TAMP** is another of ODOT's plans that prioritizes resilience. This plan details ODOT's strategy for operating, managing, and maintaining the transportation system and includes considerations of extreme weather and resilience. The TAMP also describes key strategies for managing significant risks posed by extreme weather events and climate change and includes a process for identifying and implementing resilience improvements.

There are also formal ODOT programs that are designed to increase the resilience of the transportation system. One example of this is ODOT's **Transportation Systems Management and Operations (TSMO) Program.** The TSMO program provides a basis for statewide policy and process changes aimed at increasing the focus and execution of traffic operations to better meet future system needs. One of the objectives of this program is to increase the resilience of the transportation system to winter weather events. This objective is evaluated through resilience performance measures, such as roadway clearance time. Consistent evaluation of such measures is essential for ODOT to determine the resilience of its transportation operations to extreme weather events, which in turn helps inform future adaptation planning. These measures can also be used by ODOT to evaluate the effectiveness of any completed adaptation actions and determine whether new or revised actions should be implemented. An iterative process of evaluating impacts and adaptation efforts is essential for ODOT to ensure long-lasting resilience.

Disasters can happen at any time and these disasters will continue to affect the transportation system. This is why ODOT takes steps to prioritize resilience through its disaster recovery approach. ODOT's formal **Emergency Relief flow process** includes an Event Resiliency Assessment, which ensures that resilience is considered when repairing assets following a disaster. During the Event Resiliency Assessment, ODOT determines whether the site had a reoccurrence of a previous disaster and considers a potential betterment to increase the asset's resilience to future disasters and reduce long-term expenses. Examples of betterments include raising roadway grades, stabilizing slopes and slide areas, and lengthening or raising bridges to increase waterway openings. By prioritizing resilience within the Emergency Relief flow process, ODOT is building back better and stronger after disasters.

ODOT's Emergency Preparedness

Although ODOT is working to increase the resilience of its transportation system, emergencies can still occur at any time. A robust approach to emergency preparedness is essential for creating a transportation system that can recover quickly and efficiently when extreme events do occur. As extreme weather events become more intense under future climate change, emergency preparedness is one way that ODOT can make its system more resilient to future conditions.

ODOT's Office of Emergency Operations carries out various emergency preparedness activities and is responsible for ensuring that ODOT is ready to effectively respond to both natural and human-caused disasters. Emergency preparedness activities include conducting regular statewide training and exercises with ODOT's field operations and coordinating with Ohio fire, law enforcement, homeland security, and other response agencies to facilitate efficient resolution of transportation-related emergency issues.

ODOT is also part of the Ohio Emergency Operations Plan, which is a statewide plan for emergency management.

Resilience in Asset Management

ODOT is transitioning asset management programs to a preservation-focused approach. Rather than prioritizing assets that are in the worst condition, ODOT encourages small-scale maintenance to preserve assets that are in good condition. This strategy is a cost-effective way to maximize the service life of transportation assets and ultimately create a more resilient transportation system.

This strategy is increasingly being implemented across ODOT. For example, in 2016, the Office of Pavement Engineering (OPE) transitioned from a "worst first" approach to keeping good pavements in good condition. OPE's new Pavement Management System (PMS) was a key part of this transition. PMS provides a collection of tools to support different pavement management activities. These include the reporting of current conditions and deficiencies, predicting future pavement conditions, and estimating the remaining service life. PMS also allows OPE to test different budget scenarios and ultimately determine where to focus future maintenance efforts. This approach has been extremely successful and has both improved pavement conditions and reduced maintenance costs. For example, pavement condition has improved steadily for both interstate and non-interstate pavements, as shown in Figure 13 below.

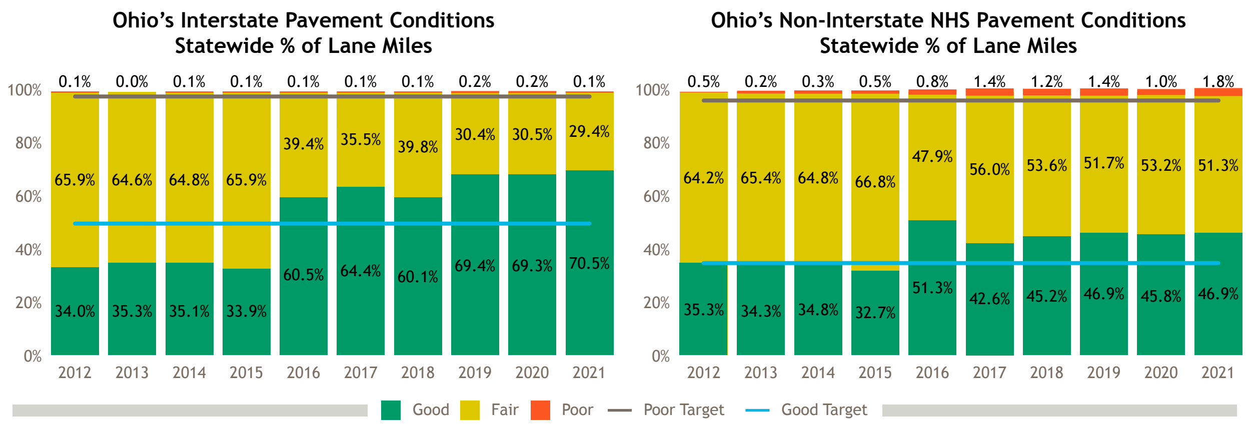


Figure 13. Ohio's Interstate (left) and non-Interstate (right) pavement conditions over time (source: *ODOT Transportation System Performance Report*, 2022).

Other engineering offices at ODOT, such as the Office of Geotechnical Engineering (OGE) and the ODOT districts, are also working to become more proactive rather than reactive in their asset management by encouraging small-scale maintenance to prevent the need for more significant repairs or complete reconstruction projects. Example maintenance activities implemented include efforts to control surface water, cleaning out ditches without compromising the slope, installing plug pile systems to stabilize embankments, and maintaining vegetation.

Resilience in Design and Engineering

In addition to incorporating resilience in strategic plans and asset management, ODOT also incorporates resilience in design to ensure that its assets are prepared for extreme weather. Some examples are updating design guidelines based on recent historical weather events; considering ways to incorporate changing climate conditions into design; researching different materials and processes to extend the useful life of assets; and integrating resilience into performance measures for operations, asset management, and preservation.

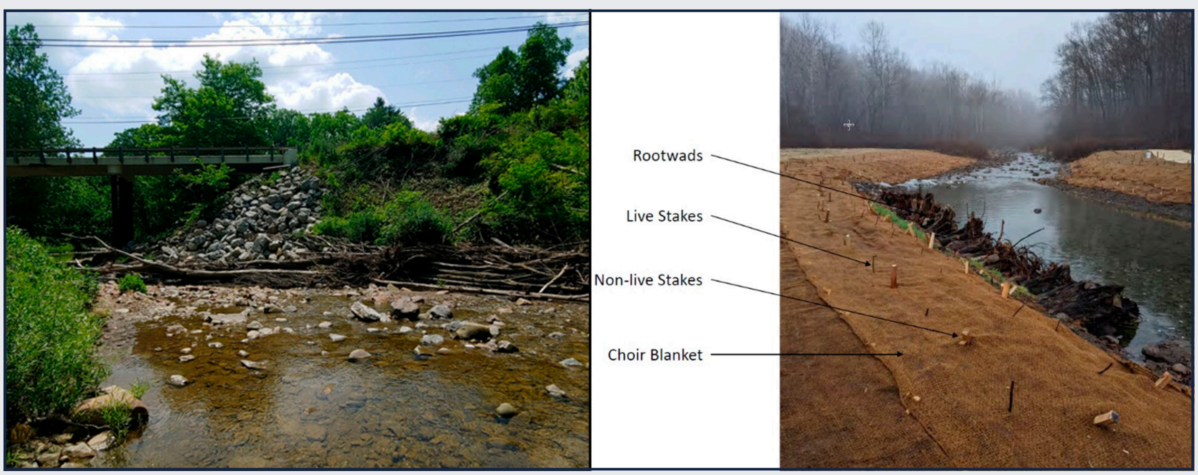
- **Risk-based design standards.** Resilience is already integrated into many of ODOT's design standards, and where possible, ODOT is exploring ways to further incorporate resilience.
    - The Office of Hydraulic Engineering (OHE) takes a risk-based approach to hydraulic design. Under this approach, design standards are tailored to correspond to the service level of the asset. More important roadways, for example, are designed to flood less frequently. By using a risk-based approach, OHE ensures that critical assets are prioritized and are made more resilient to extreme weather events and natural disasters. In addition to existing design guidelines, ODOT is working to incorporate resilience even further through updates to design standards. OHE is currently completing a research project to review climate model projections and evaluate the adequacy of current design standards to accommodate potential future changes in precipitation patterns.
    - OPE also incorporates resilience in its pavement design process. For example, ODOT's pavement design standards apply asphalt pavement binders that accommodate significantly higher temperatures than are currently experienced (or projected to occur with climate change) in Ohio. OPE also uses a chemically treated subgrade, which provides initial cost savings by reducing pavement thickness design and mitigating the need for additional undercuts during construction. In the long-term, the subgrade is more resilient due to increased stability.
- **Nature-based solutions.** Beyond its design guidelines, ODOT incorporates resilience by applying nature-based solutions and low-impact development strategies where possible. Nature-based solutions can help increase the resilience of assets to different climate hazards while simultaneously providing ecosystem benefits. One example is reconnecting natural floodplain areas and restoring river channels to reduce flooding impacts, while also improving habitats and water quality.
    - ODOT has implemented a wide range of riverine nature-based solution projects, including stream bank and riparian restoration, floodplain connectivity, natural channel design and bottomless culverts, and vegetated riprap. These projects have provided a variety of benefits, including increased flood storage; improved chemical, physical, and biological qualities; improved channel stability; and reduced erosion, in addition to helping preserve infrastructure.

- One nature-based solution approach that ODOT has successfully implemented at several locations is the Palmiter method. This method involves the use of woody debris, bank seeding, and sand and gravel bar relocation to manage bank erosion, direct river currents away from problematic erosion areas, and realign the channel flow. The Palmiter method emphasizes the use of in situ materials and minimal equipment deployment. In 2021, ODOT collaborated with the Ohio Research Institute for Transportation and the Environment to complete a research project on the efficacy of the Palmiter method in Ohio. This study concluded that the Palmiter method is a long-term, cost-effective, and sustainable option for ODOT's erosion control and stream management project needs.

Nature-Based Solutions Case Study: Furnace Run at State Route 303

In 2022, ODOT used nature-based solutions to resolve a channel migration issue along SR 303. Channel migration at the site was causing road embankment failure, bank erosion, and the accumulation of woody debris along a descending bank. ODOT implemented nature-based solutions in order to restore the stream channel to its historical alignment and ultimately reduce the risk of channel migration in the future. ODOT also conducted bank erosion repair and reconnected a portion of the surrounding floodplain.

ODOT developed a design for the project that incorporated as little rock as possible and instead included plantings and toe woods. Project construction included the implementation of a J-hook, riffle/header, and weirs to maintain the channel. ODOT also buried large boulders to ensure scour protection, repaired the roadway embankment, and planted trees along the roadway.



Furnace Run at SR 303 prior to the project (left) and the stream channel post-construction, showing rootwads, live stakes, non-live stakes, and choir blanket (right).

The project has been successful to date, and monitoring of the project will help ODOT determine the relative cost of nature-based solutions compared with gray infrastructure. Projects like this one will ultimately help incentivize nature-based solutions as a viable alternative for future ODOT projects.

- **Research into resilient materials and practices to extend asset service life.** Work is being done by ODOT to further increase resilience by researching different materials and processes to extend the useful life of assets.
  - OPE is currently working on asphalt pavement research projects to better protect the road base layer. This includes the development and testing of new mixes on different roadway segments.
  - OHE conducts extensive research on pipe materials, construction methods, and other techniques to increase the useful life of culverts.
  - OGE is conducting research into the use of remote sensing for identifying changes in potential landslide sites. These sites can then be prioritized for field inspections to identify developing landslides before they affect the transportation system.
- **Opportunistic resilience enhancements.** Finally, ODOT takes advantage of as many opportunities as possible to upgrade its infrastructure to accommodate extreme conditions. ODOT district engineers routinely identify undersized culverts and upsize them to at least meet current design standards when they need repair or replacement. District 2 (which covers part of Ohio's Lake Erie coastline) also raises bridges above the 100-year flood elevation to accommodate more extreme water levels on the lake.

Resilience in Operations and Maintenance

Frequent data collection and asset inspections ensure that ODOT is aware of the current state of most built assets to determine where to focus maintenance efforts. This is essential for ODOT to identify problems before they occur, saving costs down the line and ultimately increasing the resilience of its transportation system. Examples of resilience strategies for operations and maintenance include the following:

- **Preventive Maintenance.** Preventing problems before they occur is a critical part of ensuring a resilient transportation system that can withstand extreme conditions, reduce long-term costs to ODOT, and ensure reliable transportation. ODOT invests in maintenance through several programs, including the following:
  - The **Culvert Inspection Program** (through OHE) involves regular culvert inspections to identify those in poor condition so that they can be remediated or replaced. The frequency of inspections depends, in part, on the size of the culvert, with larger culverts being inspected more frequently. However, any culverts in poor condition are inspected annually. This program helps OHE identify potential issues before they occur and, when required, develop a cost-effective plan for rehabilitation that minimizes the impacts on the traveling public.
  - The **Bridge Inspection Program,** which is similar to the Culvert Inspection Program, helps identify scour issues early on.
  - The **Geological Site Management (GSM) Program,** which inventories landslides, rock slopes, and abandoned underground mines, helps districts address sites with geological hazard issues and identify where to focus investment. OGE is working to further develop this program with an underground mine study.

Culvert Rehabilitation Case Study

In early 2012, ODOT inspected a 108-inch span structural plate conduit under Interstate 90. The inspection revealed that the culvert was in critical condition, with invert metal loss, backfill loss, and metal plate failure. The embankment also showed signs of movement.

ODOT immediately developed a plan of action to rehabilitate the culvert that would avoid open-cut or major disruptions to traffic on Interstate 90 while still maintaining the design headwater elevation of the culvert. ODOT completed a tunnel liner replacement of the damaged portion of the culvert and concrete field paving of the entire culvert. The work was performed in 9 weeks at a cost of $345,000 with no impact on the traveling public.

Photo of the completed culvert rehabilitation project.

Without the Culvert Inspection Program, ODOT would not have discovered this critical structure before a potential collapse had occurred. This program helps save on costs for ODOT and reduces the risks to the traveling public.

  - The **Preservation Program** is responsible for providing funding for the preservation and rehabilitation of ODOT's pavements, bridges, conduits, and other assets. The Preservation Program works to maintain assets at "steady state" conditions, where preventive maintenance and regular repairs are leveraged to sustain the system conditions. This helps reduce the need for large-scale repair or replacement projects and saves money for ODOT over time.

- **TSMO Strategy.** As described earlier, ODOT's TSMO program is another important approach to promoting resilience. One of the objectives of this program is to increase the resilience of the transportation system to winter weather events. This is evaluated through resilience related performance measures, such as roadway clearance time.

- **Data Collection.** Maintaining good asset data is also a key strategy for identifying issues before they occur and determining where to focus maintenance and preservation efforts. This can help ODOT make informed decisions to help increase the resilience of its system:

  - The Asset Management program within the Office of Data Governance plays a key role in data collection for ODOT. This office uses a LiDAR system to collect detailed data on roadways and their surroundings. ODOT also has an Office of Data Governance that is responsible for consolidating data and tools to inform decision-making.

  - OHE collects and maintains data on culvert location, size, and base material. OHE has accurate data on size and material for 80% to 90% of ODOT's approximately 90,000 culverts.

Future RIP Enhancements

ODOT has identified potential actions to further enhance the RIP into the future. Potential actions include improving the risk-based vulnerability assessment methodology, updating tracking and modeling programs, increasing coordination across ODOT, providing more capacity building and training opportunities, and expanding investment.

Risk-Based Vulnerability Assessment

Potential areas of improvement in the risk-based vulnerability assessment include the following:

- Future assessments could **extend to assets beyond the floodplain.** The needs assessment is currently limited to any road and bridge assets that intersect the 100-year FEMA floodplain. Flooding and other natural hazard risks extend beyond the FEMA floodplain, especially for geohazards.
- The **evaluation of the threat likelihood** could be enhanced by using information from flood tracking (as that dataset gets a longer time series, it becomes more statistically significant) or by using hydrologic models that incorporate changing precipitation to provide additional information on the extent and depth of flooding, and likelihood of damage. Flood and damage likelihood for bridges in particular could also be improved by comparing bridge deck elevation to FEMA base flood elevation. This would provide a more accurate likelihood of flood overtopping than assigning a value to all bridges.
- The **evaluation of equity implications** currently assumes that an asset serves a disadvantaged community if it is located within that census tract. Future analyses could consider more detailed evaluation of road use by different demographic groups.
- The **redundancy analysis** currently takes a simplified approach to the road network. Future improvements could apply more detailed identification of overpasses and analysis of scenarios in which multiple segments are closed at once. Additional manual review and cleanup of the network would also help refine the results.
- Future assessments could include **monetization or valuation of critical services**. The assessment does not currently place a dollar value on whether closure of any transportation assets would result in a loss of access to critical destinations, such as fire or police stations.
- Future assessments could include **identification of constricted floodplains.** Whether a floodplain is constricted at a bridge is an indicator of the likelihood of flooding damage. However, there is not an automated way to identify constricted floodplains using geospatial analysis. Manual inspection of bridge crossings could identify constricted floodplains.

Building on Existing Resilience Practices

ODOT has also identified additional ways to build on the resilience practices already in place across the organization. Potential actions are grouped into six categories: processes, coordination, capacity building and training, practices, data and research, and investments.

Example strategies include the following:

- **Flood Incident Tracking.** ODOT could improve the level of detail recorded for flooding incidents within the existing TSMO system. Examples of additional details could include the number of lanes flooded, the duration of flooding, and the frequency of minor repairs. More detailed flood incident tracking would help ODOT and district staff better understand high-risk flood areas. Additionally, more detailed flood data would support project justification in grant applications.
- **Nature-Based Solutions.** ODOT could accelerate the application of nature-based solutions by identifying where nature-based solutions should be considered and developing a plan for piloting new practices. This strategy could also incorporate partnerships with organizations familiar with nature-based solutions to help identify and implement potential solutions.
- **Local Case Studies.** ODOT could develop a list of example resilience project types with case studies for districts and central offices to provide ideas for types of projects to consider for the improvement of resilience. Providing such case study examples with lessons learned and best practices could help facilitate future resilience projects.
- **Training on Geohazard Prevention.** ODOT could develop trainings for recommending maintenance or prevention on Tier 1 geohazard sites and how to complete planned maintenance to protect vulnerable areas. A clear set of strategies or practices for geohazard prevention would help support district staff in managing Tier 1 sites.
- **Evaluate the Cost and Benefits of Resilience Features.** ODOT could develop educational materials that demonstrate the costs and benefits of adding resilience features to infrastructure projects. This could help ODOT staff and partners understand the cost of inaction and determine when to add common resilience features.
- **Culvert Improvements**. ODOT could identify and address historically undersized culverts. Investing in culvert improvements could reduce flood risks and save costs in the long term.

Measurable Outcomes and Goals

In addition to identifying strategies to further enhance the RIP and build on existing resilience practices, ODOT will also develop additional performance measures to evaluate the resilience of its transportation operations, as well as the effectiveness of any implemented adaptation actions. ODOT recognizes that an iterative process of evaluating resilience performance measures is essential for ensuring long-lasting resilience in a climate that is constantly changing.

ODOT's TSMO program already includes resilience performance measures, such as roadway clearance time. ODOT could build on these measures to include other indicators, such as flooding and geohazard clearance time or weather-related response expenditures.

ODOT could also identify additional opportunities to measure the success of resilience efforts. For example, if ODOT develops strategies to help accelerate the implementation of nature-based solutions, a measurable outcome could be the number of projects that implement such solutions or the costs saved by incorporating these practices. ODOT's existing inventories also provide an opportunity to measure resilience. For example, as ODOT invests in culvert maintenance and improvements, the number of culverts in critical condition and the number of culverts overwhelmed or flooded during extreme events could be helpful metrics to evaluate the resilience of ODOT's culverts.

Appendix 1:

PROTECT Requirements Crosswalk

Table 3 lists the required and optional elements of a Resilience Improvement Plan (RIP) per the PROTECT program guidelines and where they are addressed in this document.

Table 3. PROTECT Requirements of a State RIP

The RIP …	Corresponding RIP Section
Shall …	
… be for the **immediate and long-range planning activities** and investments of the State or metropolitan planning organization with respect to resilience of the surface transportation system within the boundaries of the State or metropolitan planning organization, as applicable	Scope of the Resilience Improvement Plan
… demonstrate a **systemic approach** to surface transportation system resilience	Implementing a Systemic Approach to Resilience at ODOT
… be **consistent with and complementary of the State and local mitigation plans** required under section 322 of the Robert T. Stafford Disaster Relief and Emergency Assistance Act (42 U.S.C. 5165)	Connection to Existing Plans
… **include a risk-based assessment of vulnerabilities** of transportation assets and systems to current and future weather events and natural disasters, such as severe storms, flooding, drought, levee and dam failures, wildfire, rockslides, mudslides, sea level rise, extreme weather, including extreme temperatures, and earthquakes (23 U.S.C. 176(e)(2)(A-C))	Risk-Based Vulnerability Assessment
Shall, as appropriate …	
… include a description of **how the agency is prepared to respond** to the impacts of weather events, natural disasters and is prepared for changing conditions	Implementing a Systemic Approach to Resilience at ODOT
… describe the **codes, standards, and regulatory framework,** adopted and enforced by the agencies, to ensure that resilience improvements within the impacted area of proposed projects that are included in the plan	Implementing a Systemic Approach to Resilience at ODOT

The RIP …	Corresponding RIP Section
… consider the benefits of combining hard surface transportation assets, and **natural infrastructure,** through coordinated efforts by the Federal Government and the States	Implementing a Systemic Approach to Resilience at ODOT
… assess the resilience of **other community assets,** including buildings and housing, emergency management assets, and energy, water, and communication infrastructure	Scope of the Resilience Improvement Plan
… include such **other information** as the State or metropolitan planning organization considers appropriate	
May also …	
… **designate evacuation routes** and strategies, including multimodal facilities, designated with consideration for individuals without access to personal vehicles	Not applicable
… plan for **response to anticipated emergencies,** including plans for the mobility of emergency response personnel and equipment and access to emergency services, including for vulnerable or disadvantaged populations	Implementing a Systemic Approach to Resilience at ODOT
… **describe the resilience improvement policies,** including strategies, land-use and zoning changes, investments in natural infrastructure, or performance measures that will inform the transportation investment decisions of the State or metropolitan planning organization with the goal of increasing resilience	Implementing a Systemic Approach to Resilience at ODOT
… include an **investment plan** that includes a list of priority projects and describes how funds apportioned to the State under section 104(b)(8), or provided by a grant under the PROTECT program would be invested and matched, which shall not be subject to fiscal constraint requirements	Resilience Investment Plan; Appendix 4: Priority Resilience Investments
… **use science and data** and indicate the source of data and methodologies	Risk-Based Vulnerability Assessment; Appendix 3: Risk-Based Vulnerability Assessment Methodology

Appendix 2:

Existing ODOT Plans

This section describes the existing ODOT plans listed in Table 1 in the Connection to Existing Plans section. These plans identify enhanced resilience as a need or goal.

Access Ohio 2045 (Long-Range Transportation Plan)

Access Ohio 2045 is ODOT's long-range transportation plan. It provides a strategic blueprint to guide ODOT's transportation policies and investments with the ultimate goal of working toward the vision of:

"Connecting all of Ohio by a **safe**, **smart**, and **collaborative** transportation system that moves people and freight efficiently and reliably and supports **community** visions."

The plan identifies 13 strategies organized into five themes that will achieve ODOT's vision and emphasize resilience, equity, and sustainability. Several strategies are related, in part or in whole, to resilience planning and implementation. Strategy 2 calls for Ohio to proactively address transportation safety, security, and environmental risks, acknowledging that intense weather (high winds, temperature extremes, and flooding) is becoming more frequent. Other strategies focus on enhancing critical elements of the transportation system, including walking, biking, and transit networks; developing multimodal corridor plans; improving the lives of all residents; and creating jobs and supporting economic development. In addition to strategies that address risks, Strategy 11 aims to improve collaboration among the 2,500 agencies that manage Ohio's transportation system. Strategy 12 encourages transparent use and the sharing of data and information. ODOT is implementing the strategies by assembling an advisory committee, developing implementation plans for priority initiatives, and identifying progress indicators for each goal.

The RIP is consistent with the goals and strategies in *Access Ohio 2045* and will take ODOT one step closer to securing a safe, smart, and reliable transportation system.

Transportation Asset Management Plan (TAMP)

ODOT's TAMP details ODOT's strategy for operating, managing, and maintaining the transportation system, including roads, bridges, conduits, and other assets. The TAMP meets the requirements specified in 23 Code of Federal Regulations 515, which directs states to create and implement a risk-based asset management system for roads and bridges that are part of the National Highway System. The 2022 TAMP includes updates to meet requirements specified in the Bipartisan Infrastructure Law (BIL) to consider extreme weather and resilience as part of the risk management analyses in the TAMP. The TAMP summarizes current asset conditions and suggests a life cycle planning approach to reduce the risk to assets.[18] It includes an overview of ODOT's current programs and technologies that analyze road, bridge, and culvert conditions, as well as investment prioritization strategies. Section 5.9 of the TAMP summarizes ODOT's past and current efforts to enhance resilience and protect assets against extreme weather events, including findings on observed and expected climate changes from the Infrastructure Resiliency Plan and the VAST model. Additionally, the TAMP outlines plans that promote coordination between different ODOT divisions to coordinate long-term objectives. The implementation of plans

18 ODOT. 2022, July. ODOT Transportation Asset Management Plan.

is achieved by aligning ODOT's budgeting tools, optimized investment strategies, critical success factors, and the Statewide Transportation Improvement Program (STIP). Budgeting tools determine available funding, critical success factors evaluate objectives for each investment, and analysis tools select cost-effective strategies to meet objectives. The outcomes from these are then carried into the STIP. The TAMP includes key strategies for mitigating risks to assets identified as vulnerable in the Infrastructure Resiliency Plan, such as widening culverts and cleaning storm drains more frequently. Part of the annual planning process includes resiliency planning to add protective features on Federal-Aid Highway Program-eligible highways.

Transport Ohio (State Freight Plan)

Transport Ohio is ODOT's State Freight Plan. It describes Ohio's multimodal freight system, how industries use the system, system needs, issues and opportunities, and key system priorities. It is intended to guide ODOT's policy and investment decisions through fiscal year (FY) 2027 and was developed to meet existing freight plan requirements from the Fixing America's Surface Transportation Act and requirements from the 2021 BIL. A key goal in Transport Ohio is to "ensure Ohio's freight transportation system performs well, is resilient, and can enable all-important outcomes the freight transportation supports." *Transport Ohio* mentions the development of a Resiliency Plan (i.e., this document) that will identify and mitigate extreme weather and other risks to freight transportation, focusing on goods movement and critical multimodal infrastructure. Resilience is especially important for the freight system because it underpins Ohio's economy. For example, pipeline shutdowns can result in commodity shortages and higher prices, and shift the burden of transporting commodities onto other forms of transportation, such as roads. ODOT will also move toward updating design standards to harden infrastructure against extreme weather and flood events and will incorporate extreme weather considerations into Transportation Asset Management Plans.

The *Transport Ohio* Dashboard[19] contains a variety of freight system data to inform transportation plans at the state, regional, and local levels. Six Dashboard modules are available: Freight-Reliant Industries, Strategic Freight System, Truck Parking, Key Commodities and Trade, Road Needs, and Rail Needs.

The RIP fulfills *Transport Ohio's* goal of developing a Resiliency Plan and will use information from the Dashboard to identify critical risks to critical freight infrastructure and systems.

19 ODOT. n.d. Transport Ohio Dashboard. Accessed December 5, 2022. https://experience.arcgis.com/experience/865c829c347b413e9d5812f667687a42/page/Freight-Reliant-Industries/

Appendix 3:

Risk-Based Vulnerability Assessment Methodology

This appendix describes the methodology used for the risk-based vulnerability assessment in more detail.

Hazard Assumptions

Table 4 summarizes key assumptions that drive the estimated annual likelihood of closure for each asset. Closure duration and damage level assumptions by hazard are also noted in the table and factor into the consequence calculations described later in the appendix.

Table 4. Key Assumptions by Hazard

Hazard	Likelihood	Closure Duration	Damage Level
Flooding – Overtopping, no damage	For roads, assign likelihood as the highest of: a. Default by functional class design standard:[20,21] • Freeways or other multi-lane facilities with limited access: 2% annual exceedance probability (AEP) • Other highways (3,000 average daily traffic [ADT] and greater) and freeway ramps: 4% AEP • Other highways (fewer than 3,000 ADT): 10% AEP b. Watersheds[22] known to experience flooding: • Licking Watershed: 4% AEP • Hocking Watershed: 4% AEP c. Asset-specific override by district engineers based on past flooding frequency or using TSMO data For bridges, assign a likelihood of 1% regardless of the functional class.[23]	Full Closure: • Roads: 1 day • Bridges: 1 day	None

20 ODOT Location & Design Manual, Volume 2: Drainage Design. Design Annual Exceedance Probabilities from Section 1004.2. https://www.transportation.ohio.gov/working/engineering/hydraulic/location-design-vol-2/1000/1000

21 The likelihood of roadway overtopping is generally much lower than the culvert design storm. As outlined in Section 1005.1 of the ODOT Location & Design Manual, Volume 2, all highways that encroach on floodplains, bodies of water, or streams are designed to “allow conveyance of the 1% AEP storm discharge without causing significant damage to the highway, watercourse, body of water or other property.”

22 Ohio Department of Agriculture. Ohio Watersheds. Updated 2022. https://agri.ohio.gov/divisions/soil-and-water-conservation/resources/1_RWP_Landing

23 In most cases, the likelihood of flood overtopping will be lower than 1% given that bridges are designed with freeboard and some bridges are located farther above the floodplain due to site considerations. Ideally, bridge deck elevation would have been compared to the FEMA base flood elevation (BFE), but there was not sufficient data to complete this at the time of this assessment.

Hazard	Likelihood	Closure Duration	Damage Level
Flooding - Overtopping, damage	Base likelihood: See above for the annual chance of flooding. Base likelihood of flooding is then multiplied by likelihoods of damage if flooded, determined based on the following: **Roads** **Pavement condition**[24] Percentage of replacement cost if flooded:[25] • Pavement classification rating (PCR) 25-65: 2% • PCR 66-69: 1.7% • PCR 70-79: 1.4% • PCR 80-89: 1.1% • PCR 90-100: 0.8% **Bridges** Product of: a. **Constricted floodplain** (if constricted, 10% chance of damage) b. Condition: • If scour rating, waterway rating, substructure rating, OR channel rating are POOR: 10% • Otherwise: 5% c. Percentage forested (within the asset's watershed): • 0-25% - 1% • 25-50% - 2% • 50-75% - 3% • 75-100% - 5%	Full Closure: • Roads: 3.5 days • Bridges: 2.5 days Partial Closure: • Roads: 0 days • Bridges: 0 days	Repair (see the Owner Cost Calculation and Assumptions section below for a range of potential replacement costs)

24 FHWA's 2023 state-of-the-practice assessment on pavement resilience concludes that pavement response to flooding is a high-interest area of research, with some studies concluding that stronger pavements are more resistant to instances of flooding. https://www.fhwa.dot.gov/pavement/concrete/pubs/hif23006.pdf

25 Lu, Donghui, S. L. Tighe, and W. Xie. 2018. Impact of flood hazards on pavement performance. *International Journal of Pavement Engineering*. Table 6 of the paper provides damage ratios for a range of flooding return intervals across road classifications. These damage ratios range from 0.78% to 1.96%. http://dx.doi.org/10.1080/10298436.2018.1508844

Hazard	Likelihood	Closure Duration	Damage Level
Landslides	For assets included in ODOT's Landslide Inventory, assign based on the Tier: • Tier 4 (Very High) = 75% • Tier 3 (High) = 50% • Tier 2 (Moderate) = 10% • Tier 1 (Low) = 1% For assets not included in the Landslide Inventory, assign rating based on the annual probability of landslides within each landslide susceptibility zone.[26] Probability is calculated based on the historic frequency of landslides[27] per square kilometer in that zone. See Table 5. For bridges, the landslide probability was set to zero unless an inventoried landslide is located along the bridge.	Full Closure: • Roads: 3 days • Bridges: 3 days Partial Closure: • Roads: 180 days • Bridges: 180 days	Landslide repairs ($2,000 per foot)
Rockfalls	Assign a likelihood based on the Rock Slope Tier (which reflects the potential of a rockfall event to occur and the potential of rockfall or rockfall debris to affect the roadway). • Tier 4 (Very High) = 50% • Tier 3 (High) = 25% • Tier 2 (Moderate) = 10% • Tier 1 (Low) = 1% • Not rated = 0.02% These probabilities are assigned to road segments based on ODOT's Rock Slope Inventory. Any segments not aligned with a segment in the Rock Slope Inventory are assigned a 0% annual chance of rockfall because they are not near a rock slope.	Full Closure: • Roads: 2 days • Bridges: N/A Partial Closure: • Roads: 7 days • Bridges: N/A	Rockfall repairs ($2,000 per foot)

26 Based on a national landslide susceptibility map from NASA's Landslide Hazard Assessment for Situational Awareness, available for the entire continental United States at ~1 kilometer daily resolution.

27 Based on a NASA Landslide Incidents dataset. Available at Landslide Viewer (nasa.gov).

Threat Likelihood

LANDSLIDE LIKELIHOOD

Landslide likelihood was determined using a two-step approach to capture landslide probability.

NASA Cooperative Open Online Landslide Repository (COOLR) points and polygons contain citizen science data from Landslide Reporter. The Global Landslide Catalog (GLC) was developed with the goal of identifying rainfall-triggered landslide events around the world, regardless of size, impact, or location. The GLC considers all types of mass movements triggered by rainfall and some triggered by other events, which have been reported in the media, disaster databases, scientific reports, or other sources. The Sample Values command in QGIS, a free and open-source geographic information system, was used to extract the landslide susceptibility value at each historic landslide location point. The annual likelihood of a landslide occurring in each of the susceptibility areas was calculated by dividing the number of landslides in each area by 16 years (2007–2022, the years the landslide points were collected). See Table 5.

Assumption: Landslide likelihood is heavily dependent upon the COOLR dataset, which may not include all landslides that have occurred and therefore may underestimate likelihood.

Table 5 shows the landslide likelihood calculations for assets not in the ODOT Landslide Inventory.

Table 5. Assumed Likelihood of Landslides per Susceptibility Zone

Landslide Susceptibility Rating[28]	Occurrence of Landslides in Ohio, 2007-2022[29]	Total Land Area of Each Zone (in Ohio, km^2)	Annual Likelihood per km^2*
Very low	4	43,000	0.001%
Low	32	38,000	0.005%
Moderate	62	19,000	0.020%
High	2	0	Not Applicable
Very high	0	0	Not Applicable

*likelihood is the number of landslides divided by the number of years of data collection (16)

Road segments were assigned a landslide probability value by spatially joining the landslide susceptibility layer with the segments to extract the landslide susceptibility value that applies to a majority of the segment. To do this, the landslide raster layer was converted to a shapefile format.

28 Based on NASA's Landslide Hazard Assessment for Situational Awareness, available for the entire continental United States at 30 arc-second (~1 kilometer) daily resolution.

29 Based on NASA Landslide Incidents dataset. Available at Landslide Viewer (nasa.gov).

ROCKFALL LIKELIHOOD

The ODOT OGE provided locations of past rockfalls, which were matched to road segments in Excel:

https://www.transportation.ohio.gov/working/engineering/geotechnical/asset-management/rock-slope-inventory#Section400RiskScoringforInventorySites

The text box below illustrates the likelihood calculations for an illustrative road segment.

Calculation Example: Threat Likelihood

Flooding - No Damage

Highest of:

- Hocking/Licking Watershed = 0
- Default Functional Class AEP = 0.04
- District Override = 1000% (10 times/year)

Likelihood = 1000%

Flooding - Damage

Likelihood = (Likelihood of flooding) X (Likelihood of damage)

- Likelihood of flooding = 1000%
- Pavement Condition Rating = 79 → 1.4% likelihood of damage

Likelihood = 14%

Landslides

- Landslide inventory tier = not available
- Then landslide susceptibility rating = 1

Likelihood = 0.001%

Rockfalls

Rock slope inventory tier = 3

Likelihood = 25%

Owner Cost Calculation and Assumptions

Owner costs represent the cost of replacing or repairing an asset in the event of damage.

Flooding

For flooding, the owner cost is determined as a portion of the asset's total replacement cost. Replacement costs are based on asset material and geometry. The flooding replacement cost for road segments is found using the equation:

$$\text{Owner Cost}_{Road}\ (\$) = \text{Length (yd)} \times \text{Width (yd)} \times \text{Pavement Replacement Cost } (\$/yd^2)$$

where:

- **Length** is the length of the VAST segment.
- **Width** is the width of the roadway surface.
- **Pavement replacement** cost is the cost to replace each square yard of pavement. Pavement replacement costs for roads are estimated at $75/square yard for general and urban roads and $110/square yard for priority roads designated in the Road Inventory.

The replacement cost for bridges is found using the equation:

$$\text{Replacement Cost} = (\text{Deck Area } (ft^2) + \text{Approach Slab Area } (ft^2)) \times \text{Bridge Replacement Cost } (\$/ft^2)$$

where:

- **Deck area** is the area of the bridge deck.
- **Approach slab area** is the area of the approach slab, determined by multiplying the length of the slab by the width. Bridge structure data are provided in the Bridge Inventory from ODOT's Transportation Information Mapping System. However, the length of the approach slab was not included, so ODOT applied the assumptions shown in Table 6 below.
- **Bridge replacement cost** is the cost to replace each square foot of the bridge deck and approach slab. The average replacement cost for bridges is assumed to be $350/square foot.

Table 6. Bridge Approach Slab Length Assumptions

Bridge Inventory Designation	Meaning	Approach Slab Length (ft)
1	Interstate highway	30
2	US numbered highway	25
3	State highway	20
4	County highway	15
5	City street	15*
6	Federal lands road	15*
7	State lands road	15*
8	Other	15*

* Assumed.

Landslides and Rockfalls

For landslides and rockfalls, repairs include more than the cost of simply replacing the infrastructure. ODOT applied an average landslide and rock slope repair cost of $2,000 per foot of length along the roadway. Additionally, an average slide length of 203 feet was applied based on slides with lengths documented in the ODOT Landslide Inventory. For rockfalls, an average repair length of 1,000 feet was applied. The per foot repair costs were applied to the full segment length for segments shorter than the average repair length.

User Cost Calculation and Assumptions

User costs represent the total cost to all roadway or bridge users incurred while the asset is damaged or being repaired. There are two components of user cost: full closure user costs and partial closure costs. Full closure user costs result from a hazard damaging the asset to the extent that full closure is required for repairs. Partial closure costs represent the cost to users while an asset is partially closed due to traffic control and reduced speed limits in a work zone.

The total cost to users is calculated by finding the cost of taking an alternative route in the event of a full closure, the cost of impeded flow delays while a road is partially closed during and after a repair, and the additional vehicle operating cost added during delays. User costs are calculated separately for passenger vehicles and trucks due to varying economic values.

The daily user cost for either a full closure or partial closure is then multiplied by the closure durations given in Table 7 for each hazard.

Table 7. User Costs by Hazard and Asset Type

Hazard	Asset	Associated User Costs	Closure Duration
Flooding - Overtopping, no damage	Road	Full closure cost	Full closure: 1 day
	Bridge	Full closure cost	Full closure: 1 day
Flooding - Overtopping, damage	Road	Full closure cost	Full closure: 3.5 days
	Bridge	Full closure cost	Full closure: 2.5 days
Landslides	Road	Full closure cost + partial closure cost	Full closure: 3 days Partial closure: 180 days
	Bridge	Full closure cost + partial closure cost	Full closure: 3 days Partial closure: 180 days
Rockfalls	Road	Full closure cost + partial closure cost	Full closure: 2 days Partial closure: 7 days
	Bridge	Not applicable	Not applicable

The full user cost for each hazard is calculated as follows for both passenger vehicles and trucks:

$$\text{Total User Cost (\$)} = (\text{Daily User Cost of Full Closure} \times \text{Full Closure Duration}) + (\text{Daily User Cost of Partial Closure} \times \text{Partial Closure Duration})$$

Daily user cost consists of the time value of following detours, added vehicle operating costs from extra travel distance, and added highway use costs from extra travel distance.

Daily partial closure cost consists of the time value from reduced travel speeds over the length of the segment.

Time Value

Time value represents the value of time lost due to additional travel time and distance.

Time Value = Delay Time × Value of Time × Vehicles × People per Vehicle

where:

- **Delay Time**
  - For full closures: Delay time is calculated for each roadway in the Redundancy Analysis (based on the additional time needed to traverse the detour) assuming that vehicles can travel at normal speeds along the detour routes based on their functional classifications, which underestimates delay time.
  - For partial closures: Delay time is calculated assuming that vehicles can only travel at one-third the posted speed limit on the segment over the distance of the segment. Speed limit assumptions by functional classifications are provided in Table 8.
- **Value of time** is the hourly value of time, given in Table 9.
- **Vehicles** are the daily vehicle or truck volume (annual average daily traffic [AADT] or annual average daily truck traffic [AADTT]).
  - Note: If only Total AADT was available (not indicating passenger vehicles versus truck traffic), all traffic was assumed to be passenger vehicles. If both Truck and Passenger AADT were available, the time value was calculated separately for passenger vehicles and trucks and added together.
- **People per vehicle** is the average number of people in a passenger vehicle or truck, given in Table 9.

Speed limit is given by the Road Inventory for most road segments and bridges. Assets without given speed limits were assigned speed limits based on the average speed limit for each functional classification from the given speed limits for road segments. Speed limit assumptions are based on road type (functional classes 1 through 8) and not on whether the road/bridge is urban or rural. The Road Inventory uses functional classes 1 through 7 without specifying urban or rural and the Bridge Inventory uses functional classes 1 and 2 and 6 through 19 specifying rural or urban.

Table 8. Speed Limit Assumptions by Functional Classification

Functional Classification	Road Type	Assumed Average Speed Limit
1	Principal arterial interstate - Rural	65
2	Principal arterial freeway - Rural	60
3	Principal arterial other - Rural	35
4/6 (R/B)*	Minor arterial - Rural	35
5/7 (R/B)	Major collector - Rural	35
6/8	Minor collector - Rural	45
7/9 (R/B)	Local - Rural	35
11	Principal arterial interstate - Urban	65
12	Principal arterial freeway - Urban	60
14	Principal arterial other - Urban	45
16	Minor arterial - Urban	45
17	Collector - Urban	35
19	Local - Urban	35

* The functional class that comes first is designated as the road type in the Road Inventory (R) data and the functional class that comes second is designated as the road type in the Bridge Inventory (B).

Vehicle Operating Cost

Vehicle operating cost represents the additional cost of operating vehicles on an alternative route.

$$\text{Vehicle Operating Cost}_{\text{Pass Vehicles}} = \text{Reroute Mileage} \times \text{Vehicles} \times \text{Cost per Mile}$$

$$\text{Vehicle Operating Cost}_{\text{Trucks}} = \text{Reroute Mileage} \times \text{Trucks} \times \text{Cost per Mile}$$

$$\text{Total Daily Vehicle Operating Cost} = \text{Vehicle Operating Cost}_{\text{Pass Vehicles}} + \text{Vehicle Operating Cost}_{\text{Trucks}}$$

where:

- **Reroute mileage** is calculated for each roadway in the Redundancy Analysis.
- **Vehicles/Trucks** are daily vehicle or truck volume (AADT or AADTT, respectively).

- Note: If only Total AADT was available (not indicating passenger vehicle versus truck traffic), all traffic was assumed to be passenger vehicles. If both Truck and Passenger AADT were available, the time value was calculated separately for passenger vehicles and trucks and added together.

- **Cost per mile** is the cost of running a passenger vehicle or truck, provided in Table 9.

External Highway Use Cost

External highway use cost captures added societal costs per vehicle-mile traveled from increased congestion, noise, and safety, using factors from FHWA (see Table 9).

$$\text{Highway Use Cost}_{\text{Pass Vehicles}} = \text{Reroute Mileage} \times \text{Vehicles} \times \text{Cost per Mile}$$

$$\text{Highway Use Cost}_{\text{Trucks}} = \text{Reroute Mileage} \times \text{Trucks} \times \text{Cost per Mile}$$

$$\text{Total Highway Use Cost} = \text{Highway Use Cost}_{\text{Pass Vehicles}} + \text{Highway Use Cost}_{\text{Trucks}}$$

Cost Factors

The analysis uses default factors from FHWA (2023) for vehicle occupancy factors, value of time, and mileage cost, as shown in Table 9.[30]

Table 9. User Cost Factor Assumptions Variable Constants from FHWA Benefit-Cost Analysis (2023)

Variable	Passenger Vehicles	Trucks
Vehicle occupancy (people)	1.67	1.00
Value of passenger time ($/person/hour)	$18.80	$32.40
Vehicle operating costs per mile ($/mile)	$0.46	$1.01
External highway use cost ($/mile)	$0.143	$0.261
Congestion	$0.109	$0.222
Noise	$0.001	$0.021
Safety	$0.033	$0.018

Source: FHWA Benefit-Cost Analysis Guidance, 2023 Update.

30 FHWA. 2023. Benefit-Cost Analysis Guidance for Discretionary Grant Programs. https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance

System Redundancy

System redundancy is a measure of the time and distance required for vehicles to travel an alternative route. The redundancy analysis is performed for all highway segments, assuming one segment at a time is impassable.

For bridges, a redundancy analysis is not needed, as detour length is provided for bridges from the National Bridge Inventory and used in the VAST assessment.[31]

To conduct the redundancy analysis, we began by creating a 25-mile buffer around the state of Ohio to allow for rerouting outside of Ohio, if necessary (if roads are closed near the state border). Geographic information system data for the bordering state roads were obtained from the 2018 Highway Performance Monitoring System Public Release of Geospatial Data[32] and filtered to include only arterials and collectors, or roads with a functional class of 6 and above, then clipped to the buffer and joined to the ODOT Road Inventory layer.

Detour length and time for road segment closures were developed using geospatial analysis, utilizing the methods described in Redzuan et al. 2022.[33] Detour length and delay time are used to calculate user cost.

The methodology is described in detail below but essentially computes the shortest way to get from one end of a segment to the other without using the segment.

METHODOLOGY

- Create a working road network dataset that will capture all alternate routes.
- Merge the VAST segments with the buffered state Road Inventory data.
- Use the Erase function in ArcGIS to remove the pieces of Road Inventory data with VAST segments.
- Merge the two datasets together.

DATA CLEANUP

A brief visual inspection addressed the following:

- Deleting one line on double-lane highways.
- Getting rid of unnecessary lines in roundabouts.
- Getting rid of ramps that are not needed.
- Adding a few local roads that include VAST segments. These were filtered out of the working dataset because they are functional class 7.

31 Ohio DOT. Asset Reliability Final Report. p. 10.

32 Office of Highway Policy Information. 2018 HPMS Public Release of Geospatial Data. Accessed on March 20, 2023. https://www.fhwa.dot.gov/policyinformation/hpms/shapefiles.cfm

33 Redzuan, Amir Al Hamdi, Rozana Zakaria, Aznah Nor Anuar, Eeydzah Aminudin, and Norbazlan Mohd Yusof. 2022. “Road Network Vulnerability Based on Diversion Routes to Reconnect Disrupted Road Segments.” Sustainability 14, no. 4: 2244. https://doi.org/10.3390/su14042244

TOPOLOGY CLEANUP

Some segments maintained a split between intersections where the functional class switched to 7. Dangles were corrected manually using Find Dangles in ArcGIS.

Following the methods described in Redzuan et al. 2022, the shortest route was determined by simplifying the road network into a graph made of polygons representing the road system.

- Each vertex was identified either as an intersection or overpass (for our purposes, we will skip the overpass piece in the first iteration).
- The road network was turned into polygons (Figure 14). Any roads that formed a closed boundary became a polygon and the roads became edges of the polygon. Some roads were extended to close gaps.

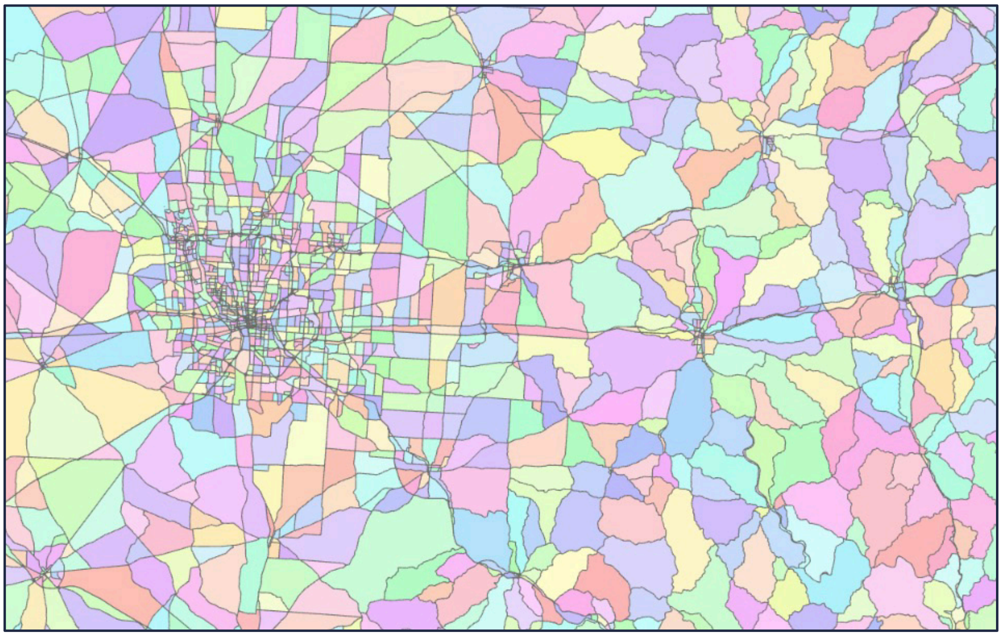


Figure 14. Polygons created from ODOT's road network.

- The shortest detour route was determined by the following:
    - Erase the VAST segments from the roads layer.
    - Merge VAST segments with the roads layers without VAST segments.
    - Recalculate segment lengths.
    - Assign all segments without a speed limit a value based on functional class.
    - Calculate the travel time for each segment, where time = distance in miles divided by speed in miles per hour.
    - Calculate the perimeter of each polygon.

- Find first and second alternative routes:
  - The first diversion route is calculated, then removed, and the second diversion route is created. The second diversion route does not include any of the paths in the first diversion route.
  - The enclosed perimeter is the original segment and the diversion routes. The total perimeter is calculated by summing the distances of both.
  - The diversion path distance for a particular segment is the polygon's perimeter minus the length of the segment.
  - The first diversion path will always be shorter than the second one. The second path is the emergency route or can be used as the primary diversion route if something such as a two-way highway interferes with the calculation of the first diversion path.
  - The ratio of the length of the original segment to the diversion route can be used to measure the route's reliability, where 1 is reliable and 0 is extremely unreliable. If both diversion routes are similar distances, then both routes are equally reliable and could be used to create a one-way direction for each path.
- Detour time will be calculated by multiplying the length of the diversion segments and the speed limit of diversion route segments.

Assumptions made in the redundancy analysis:

- The reroute distance slightly overestimates the actual travel distance because it calculates the path from one end of the vulnerable segment(s) to the other, when in reality a vehicle would turn onto a new road before reaching the other end of the segment.
- Local roads were filtered out from the initial road dataset but were added back in if they encompassed a VAST segment. Some alternative routes ended in a dead end where a local road started; these local roads were also added back in.
- There are approximately 900 VAST segments without an alternative route. These are segments that are on isolated local roads—for example, roads on South Bass or Kelleys islands in Lake Erie or dead-end streets. For these segments, the team assigned the maximum delay time and a reroute mileage of 0.

Distance to Critical Facilities

The distance to critical facilities was calculated using the Euclidean distances between the center points of each highway segment and bridge to all critical facilities (including garage or outpost, national shelter systems, hospital, fire stations, local emergency operations centers, and National Guard armories). The results were filtered to capture those facilities within a 10-mile radius of each highway segment or bridge. To eliminate the effect of outliers on normalization, all critical distance values less than the 5th percentile value are set to the 5th percentile value, and all critical distance values more than the 95th percentile value are set to the 95th percentile value. Finally, linear normalization was used to calculate the critical distance score.

Calculation Example: Consequences

Consequence = (Owner Cost) + (User Cost)

Total Owner Cost ($) =

- **Flooding:** *Length (yd) × Width (yd) × Pavement Replacement Cost ($/yd^2)*
- **Landslides/Rockfalls:** Average length (ft) × Repair Cost ($/ft)

Total User Cost ($)
= (Daily User Cost of Full Closure × Full Closure Duration)
+ (Daily User Cost of Partial Closure × Partial Closure Duration)

- **Flooding - No Damage**
  - Owner Cost = 0 (No damage to repair)
  - User Cost = $65,160
    - Daily user cost of full closure = $65,160
    - × Full closure duration = 1 day
  - **Total = $65,160**
- **Flooding - Damage**
  - Owner cost = $3,338,668
    - Length = 5,136.4 yd
    - × Width = 26.0 yd
    - × Pavement replacement cost = $75/yd2
  - User Cost = $228,060
    - Daily user cost of full closure = $65,160
    - × Full closure duration = 3.5 days
  - **Total = $3,566,728**

Calculation Example: Consequences

- **Landslides**
  - Owner cost = $406,000
    - Average landslide length = 203 ft
    - Landslide repair cost = $2,000/ft
  - User Cost = $1,402,699
    - Daily user cost of full closure = $65,160
    - × Full closure duration = 3 days
    - Daily user cost of partial closure = $6,707
    - × Partial closure duration = 180 days
  - **Total = $1,808,699**
- **Rockfalls**
  - Owner cost = $2,000,000
  - Average rockfall length = 1,000 ft
    - Rockfall repair cost = $2,000/ft
  - User Cost = $177,267
    - Daily user cost of full closure = $65,160
    - × Full closure duration = 2 days
    - Daily user cost of partial closure = $6,707
    - × Partial closure duration = 7 days
  - **Total = $2,177,267**

Calculation Example: Total Risk Value

Total Risk Value = Threat Likelihood × Consequence

Hazard	Likelihood	Consequence	Risk Value
Flooding – No Damage	1000%	$65,160	$651,599
Flooding – Damage	14%	$3,566,728	$499,342
Landslides	0.001%	$1,808,699	$18
Rockfalls	25%	$2,177,267	$544,317
Total Risk Value			**$1,695,276**

Appendix 4:

Priority Resilience Investments

Table 10. List of Priority Projects for PROTECT

Project Name	Description	Summary of PROTECT Justification
ADA SR 348-11.20	Slope stabilization from Scioto Brush Creek to SR 348 needed to prevent more damage to the roadway. Restore roadway in the area of landslide repair.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ALL SR 115-1.11	Slope stabilization and culvert replacement.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ASD SR 42-18.03 (SOLD)	This project includes construction of drilled shafts to correct and prevent roadway slippage due to landslides. It also includes replacing an existing guardrail and extending a 12-inch culvert through the drilled shaft construction.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ASD SR 58-11.80-12.00	Replace 1,000 feet of 6-inch clay pipe on the west side of the road in Sullivan (unincorporated).	There are several locations along the run of pipe that have sinkholes in them where joints have broken or separated, and much of the conduit and associated culverts are plugged with sediment.
ASD SR 511-22.18	Replace Twin Culvert Bridge, ASD SR 0511-22.18.	Replace silted-in twin corrugated metal pipe with a new culvert and upsize it.
ATB SR 531-13.20	Coastal erosion repairs to several locations along ATB SR 531 in Ashtabula County.	SR 531 along Lake Erie has been closed for long periods (e.g., 3 months) due to the erosion of the roadway embankment of SR 531 causing a safety concern. Proposed revetment project to stabilize the embankment of SR 531 from falling into Lake Erie. In one area, erosion has already taken a house and garage.
ATH SR 13-0.46	Slope stabilization project, ATH SR 13 at 0.46. Work includes a drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
ATH SR 13-13.71	ATH SR 13 at 13.71, rockfall hazard remediation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ATH SR 56-1.00	Flooding mitigation.	Flooded 43 times in the past 6 years. Roadway could be closed for multiple days per event (as much as a week).
ATH SR 144-6.04	Slope stabilization project, ATH SR 144 at 6.04.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ATH SR 144-12.08	Slope stabilization, ATH SR 144 at 12.08.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ATH SR 329-1.00 (SOLD)	Slope stabilization project, ATH SR 329 at 1.00. Geologic Site Management (GSM) funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ATH SR 690-0.42 (ENCUMBERED)	Slope stabilization project, ATH SR 690 at 0.42. Work includes a drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ATH SR 690-4.21	Slope stabilization project, ATH SR 690 at 4.21. Work includes a drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
Athens County Garage	Slope stabilization project located at the Athens County Garage.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
AUG SR 364-11.62	Embankment repair due to pavement settlement in the northbound lanes of SR 364.	Settlement is affecting the northbound travel lane of SR 364. Pavement repairs done in 2022. Maintaining SR 364 is of importance due to being the east embankment of Grand Lake St. Marys and being the route taken by lake traffic.
BEL SR 7-18.59	Mine outfall remediation along SR 7 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 9-18.70 (SOLD)	Slope stabilization by slope reconstruction and pavement repairs on SR 9 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 147-2.89	Slope stabilization on SR 147 in Belmont County by drilled shaft wall along the eastbound lane with associated culvert and pavement repairs. The design for BEL SR 379-2.26 (PID 114189) is included under this PID.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 147-24.65/25.88	Slope stabilization at two locations on SR 147 at 24.65 and 25.88 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 148-8.45	Slope stabilization along the North Fork of Captina Creek.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 148-20.21	Slope stabilization and erosion mitigation along SR 148 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 149-5.66	Scour countermeasures.	Bridge scour (SFN 0706736) along both piers has exposed drilled shafts. Project will involve scour countermeasures.

Project Name	Description	Summary of PROTECT Justification
BEL SR 149-10.82	Scour countermeasures.	Bridge scour (SFN 0704938) along abutment footing. Project will involve scour countermeasures.
BEL SR 149-12.25	Creek bank mitigation along McMahon Creek to prevent erosion of the roadway embankment.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 149-15.10	Slope stabilization along SR 149 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL US 250-6.45 (SOLD)	Slope stabilization along the south slope of US 250.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL US 250-7.90 (SOLD)	Slope stabilization on US 250 in Belmont County by drilled shaft retaining wall and associated pavement repair.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL US 250-8.15	Slope stabilization along US 250 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL US 250-9.08	Slope stabilization along US 250 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 379-1.10	Pavement patching for slope stabilization.	Tier 3 landslide (LS00008814) has caused pavement failures in the southbound lane, requiring pavement patching to keep the lane open.

Project Name	Description	Summary of PROTECT Justification
BEL SR 379-2.26	Slope stabilization on SR 379 at 2.26 in Belmont County by drilled shaft retaining wall along the southbound lane with associated culvert and pavement repairs.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 379-6.34	Slope stabilization on SR 379 at 6.34 in Belmont County by drilled shaft retaining wall and associated pavement repairs.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 647-3.29	Slope stabilization.	Tier 3 landslide (LS00004895) has progressed up the slope toward the roadway. A repair made prior to roadway failure is preferred.
BEL SR 800-1.60 (SOLD)	Slope stabilization by drilled shaft retaining wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BEL SR 800-13.15 (SOLD)	Slope stabilization by excavation and slope reconstruction.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BRO US 52-10.25	Slope stabilization project on US 52 in Brown County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BRO US 52-22.27	Slope stabilization on US 52 in Brown County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BUT SR 4-14.9	Flood mitigation.	Slide location.

Project Name	Description	Summary of PROTECT Justification
BUT SR 26-2.92	Slope stabilization.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BUT US 27-5.15	Flood mitigation.	Flooding location.
BUT SR 126-2.92	County-dumped rock.	Slide location.
BUT SR 126-9.5	Slope stabilization.	Slide location.
BUT SR 126-10.7	Slope stabilization.	Slide location.
BUT SR 128-6.62	Slope stabilization.	Slide location.
BUT SR 128-6.62	Resurfacing portion of SR 128 in Butler and Hamilton counties.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
BUT SR 129-9.1	Flood mitigation.	Flooding location.
CLE US 50-2.25	Improve the hydraulic channel at the CLE US 50-0225 bridge to minimize flooding and roadway overtopping caused by sedimentation/aggradation within the channel and protect the bridge and embankment from scour. This is a candidate project for the PROTECT funds to improve the resilience of the road from future flooding/overtopping that has been occurring with increased frequency and severity.	The bridge has been overtopped eight times in the past 21 years. Flooding is due to aggradation/sedimentation in the stream in combination with debris accumulation. Besides maintenance for debris removal, the 2018 storm resulted in damage to the road surface and flood damage to a local business. Clearance under the bridge is down to 18 inches and easily clogs with debris. Temporary right-of-way purchase is necessary to clean channel for 800 feet. Overtopping results in damage to the road, damage to local businesses, and is a danger to motorists.
CLE US 50-2.63	Replace failed retaining wall along US 50. The current wall shows signs of distress and movement. Wall failure will lead to roadway closure.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CLE US 50-7.1	Slope stabilization.	Slide location.
CLE SR 126-0.35	County-dumped rock.	Slide location.

Project Name	Description	Summary of PROTECT Justification
CLE SR 222-15.00	Flood mitigation.	Flooding location.
CLE SR 232-2.0	Slope stabilization.	Slide location.
CLE IR 275-1.48	Repair of landslide approximately 400 feet long on the downslope of Ramp C at Ward's Corner. Slide has caused damage to guardrail and lighting in the project area.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CLE/HAM-Trees	Contract to remove dead/downed trees.	Maintenance project that reduces the impacts of severe weather events.
COL SR 644-8.02	Slope repair and culvert replacement on SR 644 in Columbiana County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
COL SR 172-10.75	Replacement of culvert under SR 172 in Columbiana County that carries flow, which leads to Guilford Lake.	Complete culvert replacement (CFN 1823786, GA=3). Size of new structure to be increased due to hydraulic requirements.
COS SR 16-1.44	Flood mitigation.	Roadway flooding occurs at this location because the outflow from a 10-foot x 8-foot box culvert bridge originally intended to outlet southeast now flows westward to a series of ditches and conduits that cannot handle the flow. SR 16 is part of the Strategic Transportation System from Access Ohio 2045. The corridor provides a key link between IR 77 and Central Ohio for freight movements and regional transportation. Due to minimal detour options, closure of the roadway due to flooding causes an increased impact on the local roadway network.
COS SR 93-0.75	Geotechnical review and study of landslide location in Coshocton County along SR 93.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
COS SR 60-25.60	Embankment stabilization to correct ditch/slope failure.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CRA SR 19-21.55	Flood mitigation.	SR 19 floods a few times per year, causing the road to be closed and traffic detoured.
CUY IR 480-15.57 Drainage	Replace the barrier wall between the Ramp W-N bridge and the CSX bridge along eastbound IR 480 with a taller wall. Work will include adding catch basins behind the wall and installing a new 72-inch pipe from the catch basins to the 72-inch conduit beneath IR 480 east of the CSX bridge and repairing the slope protection at the adjacent bridges.	Flood prevention, slope erosion along mainline IR 480.
CUY IR 480-18.27 (SOLD)	Stabilize the slope along IR 480 westbound ramp to IR 77 northbound in the City of Independence.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CUY IR 480-21.38 (ENCUMBERED)	2021 GSM Project: Stabilize the slope at the northeast corner of the IR 480 bridge over Norfolk Southern railroad tracks and a major commercial drive in the City of Maple Heights.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CUY IR 480-21.61/ RAMP SLOPE	Repair slide damage along the south side of the IR 480 eastbound ramp to IR 77 southbound in Independence.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
CUY IR 480-21.61/ RAMP SLOPE	Stabilize the slope along the IR 480 ramp to Broadway Avenue in Garfield Heights. Ramp RA18272.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
CUY IR 90-CHEERS-Breakwater	Breakwater off the shore of Lake Erie to reduce high wave action against IR-90 in the City of Cleveland.	This project will construct a ~1,000 linear foot offshore breakwater structure, so that under frequent storm conditions (50% annual chance/2-year), overtopping rates will be low enough to keep IR-90 open if advisories are issued for motorists.
DEF US 24-4.73/4.84	Slope stabilization. Bench embankment, flatten slopes, and install plate piles along the westbound lanes, just southwest of the Switzer Road structure.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
DEF SR 15-12.86 (SOLD)	Repair slide area with plate piles.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
DEF US 24-1.63/2.91	Reconstruction of embankment by benching along US 24. Two locations of work include near Baltimore Street and Krouse Road. Flatten the fill slopes as much as possible and install curb and stormwater inlets to control surface runoff as needed.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
DEF SR 111-0.38-0.47	Slope stabilization.	Slope failure along the Auglaize River. The river is eroding the bank, causing "caverns" below the water line and jeopardizing the roadway above.
DEF SR 66-4.25/4.33	Slope stabilization.	Slope failure.
DEL SR 315-5.06	Improve SR 315 at various locations (between Hyatt's Road and Bunty Station Road) by resurfacing, installing roadside ditches, replacing culverts, and building retaining walls. Purchase of right-of-way will be necessary.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
DEL SR 37-6.5-6.73	Flooding mitigation.	Storm sewer is undersized. Water overtops and approaches traveling motorists.

Project Name	Description	Summary of PROTECT Justification
District-Wide PE/DD	Task Order PID for General Engineering FY 2023/2024 for D10.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ERI SR 113-5.85 (SOLD)	The existing slope is approximately 20 feet high and the embankment along the road is failing. There are numerous cracks that are starting to appear in the shoulder and the toe of the slope is starting to bulge in several areas. The failure is being attributed to both poor embankment materials and drainage over the side of the slope. The permanent repair would consist of constructing a drilled shaft secant wall behind the existing guardrail. Drainage channels or catch basins will also be needed in order to direct runoff to the bottom of the slope.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ERI SR 4-0.23 (SOLD)	Recently notified by Erie County about a slip that is occurring on SR 4 in the southwest corner of the embankment leading to the Norfolk Southern Yard overpass. Project includes the construction of drilled shafts behind the existing guardrail to correct and prevent roadway slippage due to embankment failure. The paved shoulder and curb and gutter will be restored in the slip area.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ERI SR 2-21.23	Partial culvert replacement and lining (ERI SR 0002 21.23).	Culvert could not be inspected as Lake Erie levels put it under water. Shoulder embankment has been undermined. Near a nature preserve.
ERI SR 269-11.80	Slope stabilization.	Existing embankment on the east side of SR 269 is sliding into the pond on the east side of the road.

Project Name	Description	Summary of PROTECT Justification
ERI SR 269-12.17	Flood mitigation.	Roadway tends to flood just south of SR 2. Water gets to 13 inches deep at the deepest point. This location is in flood zone AE.
ERI SR 4-0.23/0.45	Recently notified by Erie County about a slip that is occurring on SR 4 in the southwest corner of the embankment leading to the Norfolk Southern Yard overpass. Project includes the construction of drilled shafts behind the existing guardrail to correct and prevent roadway slippage due to embankment failure. The paved shoulder and curb and gutter will be restored in the slip area.	Slips are occurring on the west side of the embankment on SR 4 approaches to the bridge over Norfolk Southern railroad. Pavement and curb have separated from the shoulder and are sliding down the slope. The southbound right lane is closed and the portable concrete barrier is in place on the southwest corner. The northwest corner had some plug piles installed by purchase contract recently; however, there is more to do there as well.
ERI SR 4-0.45	Recently notified by Erie County about a slip that is occurring on SR 4 in the northwest corner of the embankment leading to the Norfolk Southern Yard overpass due to excavation along the toe of the slope. Project includes the construction of steel plate piles installed behind the existing guardrail to correct and prevent roadway slippage due to embankment failure. The paved shoulder and curb and gutter will be restored in the slip area.	Slips are occurring on the west side of the embankment on SR 4 approaches to the bridge over Norfolk Southern railroad. Pavement and curb have separated from the shoulder and are sliding down the slope. The southbound right lane is closed and the portable concrete barrier is in place on the southwest corner. The northwest corner had some plug piles installed by purchase contract recently; however, there is more to do there as well.
FAI SR 37-27.47 (SOLD)	Installation of drilled shaft plug wall to stabilize southern roadway slope along Rush Creek.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
FRA SR 3-20.96	Upsize storm sewer.	The storm sewer is undersized. In today’s environment, the magnitude of water trying to flow through this pipe is more than what the pipe can handle.
FRA US 62-5.0	Upsize storm sewer.	The storm sewer is undersized. In today’s environment, the magnitude of water trying to flow through this pipe is more than what the pipe can handle.

Project Name	Description	Summary of PROTECT Justification
FRA IR 70-15.00-15.50	Roadway flooding	Flooding events create full or partial closures of I-70 at and near the SR-79 interchange, adjacent to Buckeye Lake.
FRA SR 665-6.5	Culvert and storm sewer work on FRA SR 665 storm sewer upgrades and culvert rehabilitation. (There is the potential for temporary right-of-way.)	The storm sewer is undersized. In today's environment, the magnitude of water trying to flow through this pipe is more than what the pipe can handle.
FRA SR 674-2.48	Full replacement of bridge FRA SR 674-2.48 over Little Walnut Creek.	Scour undermining the abutments.
FUL US 20A-2.53	Flood mitigation.	US 20A at the Tiffin River (MP2.53). Water comes over the road about 200 feet west of the bridge and when it gets high enough, it will wash the berm out on the south side of the road when it is receding.
FUL US 20A-19.2	Subsidence mitigation.	North side of the road, west of Delta. Guardrail has sunk below the pavement level.
GAL SR 7-10.20	Two-lane resurfacing project using an asphalt overlay, GAL SR 7 at 10.20-19.39.	Flooded four times in the past 6 years. Roadway is typically closed for 1 or 2 days per flood. Caused by the Ohio River flood event.
GAL SR 7-19.97	Slope stabilization project, GAL SR 7 at 19.97. Work also includes a culvert replacement on GAL SR 7 at 20.03, CFN 1981270.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GAL SR 7-29.99 (SOLD)	Slope stabilization project, GAL SR 7 at 29.99. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GAL SR 160-14.33	GAL SR 160 at 14.33 750-foot rockfall hazard remediation; 450-foot slope stabilization with drilled shaft wall beginning at 14.38.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
GAL SR 553-0.61 (SOLD)	Slope stabilization project using excavation/dump rock, GAL SR 553 at 0.61-0.62.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GAL SR 554-6.61	Two-lane resurfacing project using an asphalt overlay treatment, GAL SR 554 at 6.61-20.23.	Flooded 41 times in the past 6 years. Roadway is typically closed for 1 day per flood.
GEA SR 87-0.92	FY 2013 Maintenance Project: Slope repair along westbound SR 87 (Kinsman Road) approximately at mile marker 0.92 in Russell Township. Work to be done utilizing ODOT forces and equipment with project funds to supply materials.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GEA SR 87-12.13 Replace	Replace the slab structure over Hopson's Creek located west of SR 168 in Burton Township.	Flood prevention.
GEA US 322-3.71	GSM slope repair project to stabilize the slope along the north side of Mayfield Road (US 322) located 0.11 miles east of Rockhaven Road in Munson Township. Design of this project is to be performed under PID 112663.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GEA US 322/44-8.63-14.93	Replace the Mayfield Road (US 322) drainage structure over Beaver Creek located east of Auburn Road in Munson Township and the Ravenna Road (SR 44) drainage structure located south of Bass Lake Road in Munson Township and the City of Chardon.	Flood prevention.
GEA US 322-5.76 Slope	GSM slope repair project to stabilize the slope along the north side of Mayfield Road (US 322) located 0.11 mile east of Rockhaven Road in Munson Township. Design of this project is to be performed under PID 112663.	Slope stabilization.

Project Name	Description	Summary of PROTECT Justification
GRE SR 725-2.39	Flood mitigation.	Flooding location.
GUE US 22-14.57 (SOLD)	Slope repair by installing drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GUE SR 146-29.42	Rehabilitation of rock slope and improvement of roadside ditch.	District 5 has prepared a list of rock slope sites on Tier 3 that have not been remediated previously. District 5 intends to improve these sites with additional side benefits—improved sight distance and safety. Two deficiency criteria were utilized in deciding the site for improvement. Ditch effectiveness for rock catchment and (Horizontal) Percentage Decision Sight Distance. For this location, these numbers are 38% and 12%, respectively.
GUE SR 265-5.68 (SOLD)	Reconstruction of existing SR 265 roadway with improved subgrade stabilization with geosynthetic products, granular materials, drainage, and full-depth pavement replacement.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
GUE SR 761-0.43	Repair and stabilization of failed embankment.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAM SR 32-6.82 (SOLD)	Improve the SR 32 and 8 Mile Road intersection by installing a signalized Green T intersection and improving the profile grade on 8 Mile Road.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAM US 52-36.85	Slope stabilization on two locations on US 52 in Hamilton County along the Ohio River.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
HAM IR 74-14.20	This slide is located along the south side of IR 74 within a fill section just prior to the eastbound IR 74 North Bend Road exit ramp. Oversaturated fill, in combination with a stream at the toe of the slope, resulted in failure of the embankment.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAM IR 75-5.53	Repair of landslide along the west side of IR 75 southbound, off the edge of the pavement. Current proposed repair is a buried drilled shaft wall.	This project has been identified as a priority through ODOT’s Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAM US 50-6.33	Shady Lane.	Slide location.
HAM IR 74-7.67	Near SR 128 exit.	Slide location.
HAM IR 75-3.28	Hopple Street.	Slide location.
HAM IR 75-6.35	Northbound IR 75 exit to Mitchell, near the cemetery. Above the existing retaining wall.	Slide location.
HAM IR 75-8.91 Pump	Flood mitigation.	Flooding location.
HAM IR 275-7.81	Near IR 74/275 interchange.	Slide location.
HAM IR 275-29.3	Westbound near southbound IR 74.	Slide location.
HAM IR 275-39.02	Northbound IR 275. Area was recently foam injected.	Slide location.
HAM IR 275-39.08	Southbound IR 275 between Birney Lane and US 52.	Slide location.
HAM IR 275R-5.28	Between the Kilby Road and Morgan Road overpass.	Slide location.
HAM-FWW Pump O&M	Flood mitigation.	Flooding location.

Project Name	Description	Summary of PROTECT Justification
HAN SR 568-3.1	Flood mitigation: Add a redundant route.	568 overtops due to the flooding of Blanchard River. Located in a regulated floodplain. Closes, on average, 2 to 3 times per year. Would add a redundant route into and out of Findlay during high-water events.
HAS US 22-3.59 (SOLD)	Slope stabilization along the north slope of US 22.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAS US 22-5.45	Slope stabilization on US 22 in Harrison County by drilled shaft retaining wall on the eastbound lane.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAS SR 212-0.44	Culvert replacement under SR 212 in Harrison County.	Complete culvert replacement (CFN 1985855, GA=3). Size of new structure to be increased due to hydraulic requirements.
HAS US 250-02.14	Scour countermeasures.	Bridge scour (SFN 3401537) along creek at the bottom of pile encasement. Project will involve scour countermeasures.
HAS SR 342-3.84	Slope stabilization under bridge (SFN 3402797) that carries SR 342 over Stillwater Creek.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HAS SR 646-1.92	Slope stabilization on SR 646 in Harrison County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HIG 28-10.20	Culvert and stormwater system improvements.	Culvert/storm sewer system needs improvement because the structure is starting to fail.
HIG SR 134-4.77	Culvert and stormwater system improvements.	Culvert/storm sewer system needs improvement because the structure is starting to fail.

Project Name	Description	Summary of PROTECT Justification
HOC SR 664-15.87	Two-lane resurfacing project using an asphalt overlay treatment. HOC SR 664 at 15.88-24.46. Project includes drainage and profile correction at HOC SR 664 at 23.19 using PROTECT funds.	Flooded three times in the past 6 years. Roadway is typically closed for 1 day per flood.
HOC SR 678-2.10 (SOLD)	Slope stabilization project, HOC SR 678 at 2.10-2.13. Work includes dig out and replace with dump rock.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HOL SR 520-11.80/12.91 (SOLD)	Slope stabilization at two locations on SR 520 at 11.80 and 12.91.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
HUR SR 103-2.08/2.40	Flood mitigation, pipe replacement.	Replace 1,650 feet of existing 12-inch clay pipe that is infiltrated by tree roots in Celeryville (unincorporated). Roadway floods periodically and is being pumped by a local church. A sinkhole was also found at this location in 2020.
HUR US 224-7.56/6.81	Drainage improvements.	The existing large-diameter drainage pipe is failing in several locations and there are several sinkholes that have opened in the shoulder. This project has been designed already.
JAC US 35-15.36	Slope stabilization on US 35 in Jackson County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-9.90	Fill abandoned culvert under SR 7 and replace drainage system along Market Street (TR 1392) and northbound on-ramp (RA 41017) to reduce flooding under railroad bridge.	Market Street and the SR 7 northbound on-ramp flood several times per year. Project also includes filling an abandoned drainage structure under SR 7. Flooding is very consistent since Ohio River Pike Island Dam construction. The 96-inch corrugated metal pipe culvert (CFN 1989299) built in 1956 under SR 7, which was abandoned in 1964, is at high risk for failure.

Project Name	Description	Summary of PROTECT Justification
JEF SR 7-10.37 Ramps (SOLD)	Slope stabilization by drilled shaft retaining wall along Ramp B adjacent to railroad.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-14.32 (SOLD)	Rockfall remediation along the southbound lanes of SR 7 by major slope excavation. A Type A emergency contract (PID 113188) was utilized for maintenance of traffic and emergency cleanup. The majority of this cut is funded by FHWA emergency relief funds. A ~300-foot section of the south end of the cut is funded by GSM program funds because it is beyond what is covered by FHWA emergency relief event OH20-01DDIR #JEF-001.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-22.93	Slope stabilization between SR 7 and railroad by drilled shaft retaining wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-25.67	Mine subsidence repair to utilize progressive design-build contract. This site was previously repaired in-house in 2016. A Type A emergency contract was used for exploration and Type A pavement repair in 2019 (PID 109848).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-25.67	Mine outfall remediation along SR 7 in Belmont County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 7-26.49	Scour countermeasures.	Bridge scour (SFN 4101359) along Pier 2 has exposed footing. Project will involve scour countermeasures.

Project Name	Description	Summary of PROTECT Justification
JEF SR 7-32.94	Rockfall maintenance by catchment cleanup and rockfall barrier replacement.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF US 22-6.97	Drilled shaft retaining wall for scour repair.	Bridge scour (SFN 4101839, 4101863) along Pier 4. Project will involve drilled shaft retaining wall for scour repair. Along with preventing further scour, this wall will act as an access platform for future bridge rehabilitation projects.
JEF SR 164-8.80 (SOLD)	Slope stabilization on SR 164 in Jefferson County by drilled shaft retaining wall and associated pavement repairs.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 164-9.40 (SOLD)	Mine subsidence pavement repair.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 213-4.22	Slope stabilization on SR 213 in Jefferson County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
JEF SR 213-4.57	Slope stabilization along SR 213 in Jefferson County. Currently considering walls or a short realignment as alternatives.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
JEF SR 213-18.16	Rock slope repair by scaling, slope drape installation, catchment cleanup, and rockfall barrier replacement. Originally included in 2021 GSM Stimulus program (CRRSAA, C257), but was removed from the program. A more robust repair is recommended, which will include right-of-way and environmental scope, preventing it from meeting the stimulus timeline requirements.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
KNO SR 229-14.80 (SOLD)	Slope restoration project to retard the ongoing erosion of the Kokosing riverbank.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAK SR 44-6.05/174-1.34/306-4.66 (SOLD)	A GSM-funded project to repair the slope failures on SR 44 (LAK SR 44-6.05) in Painesville Township, SR 174 (LAK SR 174-1.34) in Willoughby Hills, and SR 306 (LAK SR 306-4.66) in Willoughby/Kirtland. Work includes minor slope repair, guardrail replacement, and minimal pavement.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAK SR 84-19.01 Wall	Extend the forward wingwall of the East Walnut Avenue (SR 84) bridge over the Grand River to stop erosion to the slope along SR 84 and protect the bridge abutment.	Prevent slope erosion.
LAK SR 84-30.86	Replace the 11-foot South Ridge Road (SR 84) bridge (CMP) over Arcola Creek in Madison Township.	Bridge improvement and flood prevention.

Project Name	Description	Summary of PROTECT Justification
LAK IR 90-2.93	GSM slope repair project to stabilize the hillside, perform channel cleanout, and restore conduit capacity along westbound IR 90 west of SOM Center Road (SR 91) in the City of Willoughby Hills. Design for LAK IR 90-12.39 (PID 111517) and GEA US 322-05.76 (PID 115992)	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAK IR 90-4.28	Slope repair on the north side of IR 90 just east of Maple Grove Road.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAK IR 90-5.57 Slide	Slope stabilization along eastbound IR 90 at the Chagrin River between River Road (SR 174) and Riverside Drive in Waite Hill and Willoughby.	Slope stabilization. I-90 is a major east-west corridor.
LAK IR 90-12.39	GSM slope stabilization project along the westbound entrance ramp from the rest area in Concord Township. Design of this project is to be performed under PID 112663.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAK SR 283-4.58 Replace	Replace the Lakeshore Boulevard (SR 283) bridge over the Chagrin River overflow channel located west of the Chagrin River in Eastlake.	Flood prevention.
LAW SR 7-4.10	Rockfall remediation project on SR 7 in Lawrence County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAW US 52-7.44	Rockfall remediation on LAW US 52 at 7.44.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
LAW US 52-17.57 (SOLD)	Repair of landslide along the eastern embankment of US 52 in Lawrence County and restore pavement in work area.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAW US 52-22.09 (ENCUMBERED)	Slope stabilization in Lawrence County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAW SR 775-5.77	Slope stabilization project on SR 775 in Lawrence County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAW SR 775-8.93	Rockfall remediation on LAW SR 775 at 8.93.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LAW SR 775-19.50	Slope stabilization project on SR 775 in Lawrence County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
LIC IR 70/79	Flood remediation at LIC IR 70/79.	This location experiences repeat flooding. A planning assessment is currently underway but additional investment will be needed to support subsequent preliminary engineering, design, and construction.
LOG US 33-20.86	Slope repair.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
LOR SR 18-8.40/8.67	Flood mitigation.	SR 18 east of the Village of Wellington has a section that is prone to roadway flooding during heavy rains. When flooding occurs, it could take several days for the water to recede.
Low Drill	Flood mitigation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MAD US 40-0.62	Flood mitigation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MAH SR 14-9.18	Slope repair along MAH SR 14.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MAR SR 4-6.0	Culvert and storm sewer work on MAR SR 4 storm sewer upgrades and culvert rehabilitation. (There is the potential for temporary right-of-way.)	Storm sewer is undersized. In today's environment, the magnitude of water trying to flow through this pipe is more than what the pipe can handle.
MAR SR 61/229	Channel easement.	Channel is not being maintained downstream from two structures near this intersection and would like to get a channel easement to maintain the flow and ensure that flooding does not occur at this intersection.
MAR SR 95-9.2	Flood mitigation.	Water overtops the roadway in the winter months and freezes regularly, causing motorists to reroute.
MAR SR 746-3.66	Upsize storm sewer.	The storm sewer is undersized. In today's environment, the magnitude of water trying to flow through this pipe is more than what the pipe can handle.

Project Name	Description	Summary of PROTECT Justification
MED SR 57-18.66	Flood mitigation.	North of Lester Road under railroad bridges. Usually closes whenever heavy rains occur. Have been dealing with this for many years.
MED IR 76-8.05	Drainage improvements.	Improve drainage on the IR 76 eastbound acceleration lane from SR 57 because spring drain water is rising up through the pavement. Attempted to fix this with a recent project; however, the problem is persisting.
MED SR 94-7.64/7.91	Asphalt concrete overlay with repairs, LOR SR 162-0.00 to 8.743; MED SR 162-24.168 to 26.751. asphalt concrete overlay without repairs. MED SR 94-7.64 to 7.91. Bridge rehabilitation to be included.	Replace existing conduit and curb on SR 94 south of Sharon Circle. Existing stormwater runoff erodes graded shoulder and creates flooding on private property.
MED IR 271-5.64	Slope stabilization.	Slip on southbound embankment, entire slope collapsing up to the guardrail posts. Approximately 100 feet.
MEG SR 124-62.80 (SOLD)	Slope stabilization project, MEG SR 124 at 62.80. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG SR 143-9.20 (SOLD)	Slope stabilization project, MEG SR 143 at 9.20. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG SR 143 11.570 (ENCUMBERED)	Slope stabilization project, MEG SR 143 at 11.57. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG SR 681-3.21 (ENCUMBERED)	Landslide remediation of seven landslides between MEG SR 681 at 3.21 and 3.80. Work includes a culvert replacement on MEG SR 681 at 3.87 (CFN 1881079).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
MEG SR 681-3.32	Landslide remediation of seven landslides between MEG SR 681 at 3.21 and 3.80. Work includes a culvert replacement on MEG SR 681 at 3.87 (CFN 1881079).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG SR 681-6.24	Slope stabilization project using dump rock, MEG SR 681 at 6.24.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG US 33-4.45	MEG US 33 at 4.45 800-foot rockfall hazard remediation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG US 33-6.30	MEG US 33 at 6.30/6.45 686-foot/438-foot rockfall hazard remediation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG US 33-7.61	Rockfall hazard remediation in the westbound lanes of MEG US 33 at 7.61-7.73 and 7.87-7.94.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MEG SR 124-9.20	Flood mitigation.	Flooded 46 times in the past 6 years. Roadway could be closed for multiple days per event (as much as a week).
MOE SR 7-7.55	Slope stabilization project, MOE SR 7 at 7.51. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MOE SR 26-14.76 (SOLD)	Slope stabilization project, MOE SR 26 at 17.76-14.80.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
MOE SR 78-8.30	Slope stabilization project, MOE SR 78 at 8.30.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MOE SR 255-16.38	Slope stabilization project, MOE SR 255 at 16.38.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MOE SR 536-5.97	MOE SR 536 at 5.97 316-foot rockfall hazard remediation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MOE SR 537-0.32	Slope stabilization, MOE SR 537 at 0.32. Work includes an asphalt overlay on MOE SR 537 at 0.00-0.66.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MOT SR 4-19.30	Repair settled pavement along mainline SR 4 and the northbound on-ramp from Stanley Avenue.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MRG SR 37-8.09	Slope stabilization project, MRG SR 37 at 8.09.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MRG SR 78-8.27	MRG SR 78 at 8.27 300-foot slope stabilization with drilled shafts.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
MRG SR 78-10.96	MRG SR 78 at 10.96 165-foot slope stabilization with drilled shafts.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MRG SR 376-1.10 (SOLD)	Slope stabilization project, MRG SR 376 and 1.10. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS SR 16-6.70	Slope stabilization along slope of SR 16.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS US 22-9.17	Flood mitigation.	Roadway flooding occurs approximately six times per year in South Zanesville at this location. District 5 anticipates that storm sewer drainage capacity increases are needed throughout this location.
MUS IR 70-7.40	Embankment slope stabilization along IR 70 at SLM 7.36 (mile marker 150.14 westbound).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS IR 70-25.90 (SOLD)	Re-establishing the stabilization of a failed embankment area on the north foreslope. Maintenance OPID assigned is 2442 (for Lo-drill plug shaft wall).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS IR 70-27.00	Mitigation of slope failure along the IR 70 eastbound exit and entrance ramps to SR 83.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
MUS SR 146/340-31.42/3.26 (SOLD)	Removal of rock from slope by excavation at the following locations: Part 1 MUS SR 146 (31.42 to 31.49) and Part 2 MUS SR 340 (3.26 to 3.32).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS SR 146-29.46	Rehabilitation of rock slope and improvement of roadside ditch.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS SR 376-5.19	Repair of embankment (cut slope) and installation of catchment ditch.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS SR 555-2.346	Installation of drilled shaft wall to address slope failure along SR 555.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS SR 83 Ramp to IR 70 Eastbound	Mitigation of slope failure along the IR 70 eastbound exit and entrance ramps to SR 83.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
MUS/COS Lo Drill	Stabilize slope at locations throughout District 5 by D5 Roadway Services.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
NOB SR 146-16.35	Slope stabilization project. NOB SR 146 at 16.35. Work includes drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
NOB SR 147-9.93	Slope stabilization project. NOB SR 147 at 9.93. Drilling for a geotechnical exploration.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
NOB SR 821-18.31	Slope stabilization project, NOB SR 821 at 18.31-18.36. GSM funds.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
OTT SR 105-4.11 (SOLD)	Cameras for the Monroe County Public Transportation building.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
OTT SR 163 24.16	Repair revetment along the north side shoreline of SR 163 (West Lakeshore Drive) in the City of Port Clinton. Perform the necessary related work. The lake continues to wash away the current revetment material. There are areas of the shoulder that are beginning to be undermined.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
OTT SR 2-17.4-18.5	Repairing the Portage River Causeway on OTT SR 2.	Repairing the Portage River Causeway on OTT SR 2.
OTT SR 2-17.57	Slope stabilization on SR 2 in Ottawa County at SLM 17.57.	The embankment is actively falling away, and the shoulder is in immediate jeopardy.
OTT SR 357 at the end	Flood mitigation.	Flooding when lake levels come up.
PAU/DEF US 24-3.83	Slope stabilization at multiple locations along US 24 by regrading the slopes and installing plate piles. Use Geohazard Maintenance purchase contract no. 059.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
PAU US 24/127-13.52/15.98 (SOLD)	Slope stabilization by reconstructing and benching the embankment along northbound US 127, the eastbound entrance ramp to US 24 (Ramp D), and along US 24 eastbound northeast of the US 127 interchange. The project will also flatten the slopes and install curb and stormwater inlets to control surface runoff.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PAU US 24-13.52 (SOLD)	Reconstruction of the embankment by benching along US 24 westbound. Flatten the fill slopes as much as possible and install a spaced drilled shaft wall. Install curb and stormwater inlets to control surface runoff as needed.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PAU SR 111-23.80-24.09	Slope stabilization.	Slope failure along the Auglaize River. The river is eroding the bank, causing "caverns" below the water line and jeopardizing the roadway above.
PAU SR 637-12.26	Slope stabilization.	Slope failure along the roadway.
PIC SR 56-20.0 to 20.8	Flood mitigation.	Creek scour leading to embankment slip. Current creek is 12 feet from the edge of the roadway and the creek is 15 feet below the pavement surface.
PIC SR 56-20.77	Slope stabilization.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIC SR 56-33.85	Scour countermeasures.	Scour undermining the piers.
PIC SR 62-6.94	Replace a culvert on SR 62 in Pickaway County. Culvert is located at SLM 6.94, which is 100 feet south of Harrisburg Road.	Culvert is undersized.

Project Name	Description	Summary of PROTECT Justification
PIC SR 762-12.78	Pavement resurfacing and repair pavement failures and slope erosion on PIC SR 762 asphalt concrete overlay without repairs PIC SR 762-11.18-14.89 (US 23 to Rickenbacker Parkway).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIC/FAI/ FRA SR 674-0.00/0.00/0.00 (SOLD)	Asphalt concrete overlay with minor pavement repair, bridge deck sealing, and upgrade guardrail as needed. Rebuilding slope, shoulder, and southbound lane at 10.32 to stabilize the slope along SR 674. PICSR 674-0.00 to 10.93 (including small portion in Fairfield County/ District 5); US 22 to Franklin County Line and FRA SR 674-0.00 to 1.91.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIK SR 335-20.78	Slope stabilization on SR 335 in Pike County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIK SR 551-1.35	Slope stabilization on SR 551 in Pike County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIK SR 772-14.10	Slope stabilization on SR 772 in Pike County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PIK SR 32-6.60	Bridge maintenance to address embankment erosion on SR 32 in Pike County.	Bridge maintenance to address embankment erosion on SR 32 in Pike County.
POR IR 76-14.89 (SOLD)	Slope stabilization along Porter Road near the POR IR 76 overpass. Work is in the northwest quadrant.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
POR IR 76-20.05 (SOLD)	Slope stabilization (benching and drainage) along IR 76 at westbound and eastbound SLM 20.05 to 20.25, in Palmyra Township, Portage County, Ohio.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PUT SR 694-0.23	Slope stabilization along SR 694 by regrading the slope and installing plate piles. Use Geohazard Maintenance purchase contract no. 059.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
PUT SR 108-2.66	Construct two plug pile walls in Miller City from SLM 2.66 to 2.84.	Slope failures keep reoccurring. Dumped rock has been placed; however, the slopes keep moving. A wall is needed as a permanent solution. Miller City cutoff ditch runs parallel to SR 108. The foreslopes of the ditch (approximately 10 feet deep) are failing. Other sections of SR 108 have already failed and walls have been built. Miller City receives $8,000 annually in gas tax revenue and has no way to fund the project. The plug pile wall is proposed as previously dumped rock continue to move. Slope failure is imminent.
PUT SR 190-5.72	Flood mitigation.	Location overtops twin culverts several times per year. Existing twin culverts need to be addressed and the roadway profile may need to be raised.
PUT SR 634-9.66-9.75	Flood and slide mitigation.	Location floods several times per year, resulting in road closures that last multiple days. High water has caused multiple slides in the foreslope.
RIC SR 13-24.96/25.00	Flood mitigation – raising embankment and bridge.	SR 13 floods at the south junction with SR 96 and needs to be closed several times throughout the year. Propose raising the embankment and bridge.
RIC IR 71-15.79	Slide mitigation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
RIC SR 96-16.58	Bridge replacement - existing prestressed box beam bridge, SFN 7005474.	Replace two span 149-6" with single span 150' steel beam bridge. Eliminate potential scour at the pier in the stream that turns 90 degrees into the bridge just upstream of the bridge. Bridge is slightly realigned along the existing roadway centerline to better fit the stream and lessen erosion on the bank below the bridge, against which the stream flows.
ROS SR 772-10.30	Slope stabilization project on SR 772 in Ross County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
ROS SR 41-8.10	Drainage improvements.	Culvert needed at location due to improper drainage from an old railroad bed.
ROS SR 772-8.76	Bridge erosion/scour remediation on SR 772 in Ross County.	Bridge erosion/scour remediation on SR 772 in Ross County.
SAN SR 19-16.985	Flood mitigation.	Flooding at curve due to the lowest point during heavy rains.
SAN SR 19-2.75	Slide mitigation.	Pavement slide southbound.
SAN SR 53-17.69	Flood mitigation.	Flooding just south of the Ottawa County line and north due to the rising water level of Muddy Creek, mainly when a northeasterly wind blows across Lake Erie into the bay.
SAN SR 523-3.123	Flood mitigation.	Flooding just east of bridge 7203284 due to the rising water level of Little Muddy Creek, mainly when a northeasterly wind blows across Lake Erie into the bay.
SCI US 52-15.24	Slope stabilization on US 52 in Scioto County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
STA IR 77-2.57	Repair 200-foot retaining wall near southbound STA IR 77 at 2.57.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
STA SR 225-2.68	Slope stabilization on STA SR 225 at 2.68.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
TRU SR 5-6.41 (RR) (SOLD)	Slope stabilization on SR 5 westbound in Trumbull County.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
TRU SR 5-11.95	Repair slide on TRU SR 5 at 11.95 (eastbound).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
TRU SR 800-0.53	Slide mitigation.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
UNI SR 245-0.48	Construct new 86-foot three-span bridge and roadway to prevent overtopping of SR 245. Project will be processed as Limited Review.	Water overtops SR 245.
VAN US 30-14.00	Slope stabilization by regrading the slope and installing plate piles. Use Geohazard Maintenance purchase contract no. 059.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.

Project Name	Description	Summary of PROTECT Justification
VAN SR 81-2.55	Reconstruct and raise profile of roadway. Replace two culverts.	Flooding events often close SR 81, cutting off access to the Village of Wilshire. Overtopping of SR 49 within the Village due to the St. Marys River closes both state routes into and out of the Village. Both state routes into and out of the Village have overtopped simultaneously, causing the Village to be isolated during flooding events.
VIN SR 328-8.97	Slope stabilization project, VIN SR 328 at 8.97.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
VIN SR 356-1.33	Slope stabilization project, VIN SR 356 at 1.33. Work includes drilled shaft wall.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
VIN US 50-8.65	Flood mitigation.	Flooded 19 times in the past 6 years. Roadway is typically closed for 1 day per flood.
VIN SR 93-7.20	Flood mitigation.	Flooded 13 times in the past 6 years. Roadway is typically closed for 1 day per flood.
VIN SR 278-0.00	Two-lane resurfacing project using an asphalt overlay treatment on HOC SR 278, 0.00-0.26; VIN SR 278, 0.00-12.99; and SR 677, 4.33-4.78.	Flooded 72 times in the past 6 years. Roadway is typically closed for 1 day per flood.
VIN SR 278-0.00	Two-lane resurfacing project using an asphalt overlay treatment on HOC SR 278, 0.00-0.26; VIN SR 278, 0.00-12.99; and SR 677, 4.33-4.78.	Flooded 37 times in the past 6 years. Roadway is typically closed for 1 day per flood.
WAR SR 123-9.93	North of Morrow.	Slide location.
WAR SR 350-8.3	Undermining of existing wall.	Slide location.
WAR SR 741: SLM 11.5-12	Elston.	Flooding location.

Project Name	Description	Summary of PROTECT Justification
WAS SR 676-2.50 (SOLD)	Slope stabilization project, WAS SR 676 at 2.50. GSM funded.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WAS SR 60-15.79	WAS SR 60 at 15.79, 200-foot slope stabilization with drilled shafts.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WAS 550-14.33	Slope stabilization project, WAS SR 550 at 14.33.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WAS SR 821-4.44	Two-lane resurfacing project using an asphalt overlay treatment, WAS SR 821 at 4.44-11.74.	Flooded five times in the past 6 years. Roadway is typically closed for 1 or 2 days per flood.
WAS SR 821-8.69	Slope stabilization project, WAS SR 821 at 8.69.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WAY SR 95-8.03	Flood mitigation.	SR 95 floods just east of Funk (unincorporated) and needs to be closed several times throughout the year. Propose raising the embankment and bridge.
WAY SR 241-1.32	FY 2025 GSM Funds Application, WAY SR 241 at 1.32.	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WAY US 250-17.27	Bridge replacement, WAY US 250 at 17.27 (over branch of Apple Creek).	Replace a three-span slab bridge in kind. Bridge will be raised slightly and will not be overtopped by the 100-year high-water event as the existing bridge is.

Project Name	Description	Summary of PROTECT Justification
WAY SR 302-3.60/4.40	Flood mitigation.	SR 302 floods and needs to be closed several times throughout the year. Propose raising the roadway and upsizing several culverts along that run.
WOO IR 475-0.39	A GSM-funded project to repair the slope on IR 475 in Wood County near the IR 75 Interchange (SLM 0.39).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WOO SR 280-5.00	WOO SR 280 southbound slope stabilization (SLM 5.00).	This project has been identified as a priority through ODOT's Geologic Site Management Program, which identifies locations across the ODOT system that need preventive maintenance investments to avoid impacts from geohazards.
WYA SR 199-2.0	Flood mitigation – redundant route.	Roadway overtops due to Sandusky River flooding. Would add a redundant route into and out of Upper Sandusky during a high-water event.
Clear sediment from obstructed culverts	Restore conduit capacity by clearing sediment from obstructed culverts.	Thousands of ODOT's ~90,000 inventoried culverts are classified as having a "partial" condition assessment, meaning that the culvert is at least partly full of sediment or water, so its exact condition cannot be ascertained. These represent culverts that are already at least partially obstructed, meaning that they are not contributing their full capacity to drain roads and neighborhoods in the event of heavy rain or flooding events. Cleaning these culverts will directly increase ODOT's capacity to drain stormwater during extreme rains or flood events and prevent roadway overtopping.
Geohazard site inspections	Geohazard site inspections.	Inspections and preventive maintenance are a critical strategy to prevent landslides or rockfalls before they happen, and to maintain the reliability of Ohio's transportation network. Prevention is significantly less expensive than repairs, which typically cost ~$2,000 per foot.
Improve mowing practices	Reduce mowing and add pollinator locations.	Maintenance practice that improves water and air quality; potentially reduces runoff.

Project Name	Description	Summary of PROTECT Justification
Increase shoulder sweep	Contract to sweep shoulder to prevent debris downstream.	Flooding: Maintenance practice that reduces the impacts of severe weather events.
Install inlet screens	Cover catch basin inlets to prevent debris.	Flooding: Management practice that reduces the impacts of severe weather events.
Underdrain outlet inspection program	Inspect underdrain outlets.	This strategy will allow ODOT to reduce pavement damage from flooding events. Underdrain outlets are another critical chokepoint for stormwater; these outlets need to be open and clear in order to drain subgrades and ensure that heavy rainfall or flooding does not damage pavements.
Upgrade culverts in poor condition	Culvert upgrades.	Culverts in poor condition create a risk for flooding and washouts across ODOT's system. Repairing these culverts (including upsizing them as needed) will prepare ODOT for a potential increase in the frequency or severity of heavy rain events.
Vegetation management	Vegetation management.	An increasing number of dead trees along Ohio roadways presents a risk to system reliability and safety. These trees are especially at-risk during winter storms or freezing rain events, which are more likely to cause the trees to fall over when road clearance resources are already occupied with snow and ice removal.

Ohio Department of
Transportation